

PRINT-FLOW-360 · RESEARCH LIBRARY

Metrics Analytics

7 research documents · each with plain-language key takeaways

Compiled 2026-06-26

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Print-Flow-360 Funnel Analytics & Event Taxonomy Design

Source: [readme/research/metrics-analytics/FUNNEL_ANALYTICS_EVENT_TAXONOMY_2026-06-16.md](#)

IN SHORT — KEY TAKEAWAYS

This document is a step-by-step plan for tracking what store owners do after they sign up — from creating their first product to receiving their first customer order. It tells the team exactly which actions to record, how to name them, and where to record them so the numbers are always reliable. Without this, you cannot see where store owners get stuck and you cannot fix what's broken.

The main parts explained simply:

- **Funnels** — A funnel is a series of steps you expect a store owner (or customer) to complete in order, for example: sign up → publish store → receive first order. You measure how many people complete each step. This shows you exactly where people stop and leave, so you know where to focus your improvements.
- **Conversion window** — The maximum amount of time you allow between the first and last step. If a store owner signs up on Monday and goes live on Thursday, you need a window of at least four days or they look like they “failed” when they didn’t. The document recommends 7–14 days for the sign-up-to-first-order journey.
- **Event naming rules** — Every action you record needs a clear, consistent name. The rule is simple: name every event as “thing_what-happened”, all lowercase, past tense — for example `order_received` or `store_published`. If names are inconsistent, your reports become impossible to trust.
- **Events vs. properties** — An “event” is what happened (`product_added`). A “property” is the detail about it (`product_type: business cards` , `price: 24.00`). Keeping these separate prevents your list of tracked actions from growing out of control.
- **Identity stitching** — Before someone signs up, they are anonymous (the system doesn’t know who they are). The moment they create an account, the system must connect their earlier anonymous steps to their new profile. If this is done wrong, one real person looks like two different people in your reports, and the numbers lie silently.
- **Server-side vs. browser-side tracking** — Tracking from the browser is easy but unreliable: browser privacy tools and ad-blockers can silently drop important events, especially orders and payments. Tracking from the server (the backend) cannot be blocked. The document says: track every money and order event from the server; let the browser handle low-importance clicks automatically.
- **Tooling choice (PostHog)** — PostHog is recommended as the analytics tool. It is free up to one million events per month, records session replays (video-like recordings of what users do), and handles funnels, feature flags, and A/B tests in one place. It can also be self-hosted for privacy control.
- **The 14-event starter plan** — Instead of tracking hundreds of actions, the document gives a focused list of 14 key events that cover the whole store-owner journey: from landing on the site, through setting up products and pricing, to publishing a store and receiving a first order. Tracking fewer, more meaningful events is far more useful than tracking everything.

What to do with this:

Add tracking for the seven most important server-side events (signup, product added, store published, order received, first order received, payment card added, subscription changed) inside the existing Laravel backend code — these already happen there, so it is less work than it sounds. Set up PostHog with the 14-event plan and build one weekly funnel: sign up → store published → first order, with a 7–14 day window. Review that single funnel every week to find the step where the most store owners drop off.

Internal analytics/instrumentation strategy doc. Researched 2026-06-16 via a multi-agent workflow (funnel-analytics, event-taxonomy, identity-stitching, server-vs-client, tooling, and starter-plan streams — each web-researched against primary sources and cross-checked). Benchmarks are flagged by confidence; **treat every vendor-blog percentage as a hypothesis to A/B-test on our own funnel, not a guarantee.** This doc is the instrumentation companion to

[readme/CONVERSION_FUNNEL_RESEARCH_2026-06-15.md](#) (which defines the AARRR strategy and the North Star: *store live + first order in 7 days*). Where that doc said “instrument the activation funnel,” this one says exactly *how*.

Executive Summary

For a low-ACV, non-technical SMB print SaaS, the single most common analytics mistake is **over-instrumentation**: a 200-event tracking plan that nobody owns, fires inconsistently, and obscures the one funnel that matters. Our buyers are print-shop owners, not a million-MAU consumer app — we have **limited eng time and a small customer base**, so signal-per-event is everything. The opinionated recommendation: **instrument a deliberately small, server-side-anchored event set (~12–15 events) that traces the real lifecycle — signup → store setup → product added → store live → first order → reorder — written in one strict `object_action` snake_case convention, governed by a single one-page tracking plan with a named owner per event, and fire every revenue/order event SERVER-SIDE so ad-blockers and browser privacy can never punch a hole in the numbers we’ll make decisions on.** Pick **PostHog** (generous free tier, bundles funnels + session replay + feature flags, self-hostable, EU/data-control friendly) as the default tool; its autocapture covers low-value clicks for free so we only hand-instrument the high-value lifecycle events. Build the funnel **strict-ordered with a generous conversion window (7–14 days)** for activation, and resist adding events until an existing one has answered a real question.

A note on the numbers. The *principles* here are well-grounded across Amplitude, Mixpanel, PostHog, and Segment primary docs. The *precise percentages* — especially ad-blocker data-loss (the widely-repeated “25–40%”) and “X% of revenue events lost” — trace to vendor blogs and are explicitly **unaudited**. Ad-blocker *adoption* (~29–33% of users) is well-measured; ad-blocker *analytics-data-loss* (20–40%) is a directional estimate. Treat tool free-tier limits as current-as-of-2026 and re-verify before committing.

1. Funnel analytics — defining, windowing, and segmenting funnels

What a funnel is. An ordered set of events a user must complete (e.g. `signup_completed` → `store_published` → `order_received`), measured as the **% of users who reach each step** within a **conversion window**. Funnels count **unique users**, not raw events — a user converts once per attempt [1][2].

Conversion window. The maximum time allowed between the first and last step. Mixpanel allows windows from a **2-second floor (strict) up to 90 days**; loose-mode windows run **1–90 days** [1][3]. The window is a *modeling choice*, not a detail: too short and you under-count slow-but-real conversions; too long and you over-credit unrelated sessions. For an activation funnel where owners may sign up Monday and go live Thursday, **a 7–14 day window matches our North Star** rather than a same-session window.

Strict vs. loose (flexible) ordering — the most consequential funnel setting.

- **Strict:** the user must do the steps in *exact* sequence with nothing in between counted out of order. Use for **linear, must-follow flows: checkout, payment, the go-live wizard**. [2][3]
- **Loose / flexible:** the user must hit all steps *in order* but may do other actions between them. Use for **exploratory journeys** (signup → eventually published → eventually first order, with browsing/editing in between) [2][3].
- Mixpanel applies a **2-second tolerance**: consecutive steps within 2s of each other are treated as interchangeable (handles events that fire near-simultaneously) [1].

Multi-path funnels. Real products have more than one route to value. The pitfall: forcing a single linear funnel through a step many users legitimately skip — it **inflates the drop-off at that step and produces a false leak** [2][4]. Fix by (a) making questionable steps optional in the funnel definition, or (b) modeling separate funnels per path and comparing.

Segmentation is where funnels earn their keep. A blended funnel hides the leak. Always break the same funnel down by **acquisition channel, persona (“what do you sell?”), device, and signup cohort** to localize *where* and *for whom* it leaks [4][5].

Common funnel pitfalls (cross-checked):

Pitfall	Why it misleads	Confidence
Requiring an unnecessary step	Users who skip it look like drop-offs; skews the whole funnel	Solid [2][4]
Wrong conversion window	Too short under-counts real conversions; too long over-credits	Solid [1][3]
Strict where loose is correct (or vice-versa)	Counts the wrong population entirely	Solid [2][3]
Identity-merge lag (anon → known)	Can create ~24h discrepancies; pre-merge cohorts look wrong	Verified [1]
Funnel vs. raw-count confusion	Funnels need sequence; “Insights”/totals count raw events — different denominators	Verified [1]
Only ever looking at the blended funnel	Masks channel/persona-specific leaks	Solid [4][5]

2. Event taxonomy design — names that won’t rot

A taxonomy is a **controlled vocabulary** for events and properties. It rots when engineers invent names ad hoc under deadline. The discipline below is the consensus across Amplitude, Mixpanel, PostHog, and Segment [6][7][8].

The `object_action` (verb_noun) convention. An event = an *object* (noun: the thing) + an *action* (verb: what happened to it). Amplitude’s house style is `button_clicked`, `checkout_completed` — **lowercase, underscores, past tense** [6]. The four naming axes to lock down once and enforce forever [7]:

1. **Casing** — pick ONE (`snake_case` recommended; survives data-warehouse export better than `camelCase`). Casing drift alone fragments metrics. [6][7]
2. **Format** — `object_action` (or `context_object_action`) so related events cluster.
3. **Tense** — **past simple** (`order_received`, not `receive_order`) — reads as a historical record.
4. **Vocabulary** — restrict verbs to an approved list. PostHog’s default set (`view`, `click`, `submit`, `create`, `add`, `update`, `delete`, `remove`, `start`, `end`, `cancel`, `fail`, `send`, `invite`, `generate`) is a strong starting point. [7]

The single most important code rule: event names are FIXED STRINGS. Never build an event name from a variable (`order_${status}`). Variable data goes in **property values**, not event names — otherwise every status becomes a new “event” and the taxonomy explodes. [7]

Events vs. properties vs. user-properties. [6]

- **Event** = the discrete action (`product_added`).
- **Event property** = context of *that occurrence* (`product_type: "business_cards"`, `price_set: 24.00`, `source: "demo_seed"`).
- **User property** = persistent attribute of the *person/store* (`plan: "growth"`, `store_vertical: "print_shop"`, `is_b2b: true`).

- **Rule of thumb:** if you're tempted to make `product_added_business_cards` and `product_added_flyers` separate events, that's two **property values** of one event. Properties prevent sprawl. [6][7]

Property naming. `object_adjective` (`product_type` , `order_total`); prefix booleans `is_` / `has_` ; suffix dates `_date` / `_timestamp` . Keep property *types* stable — sending the same property as a string then a number silently breaks aggregations. [7]

Sizing guardrails (Amplitude guideline, treat as functional, not law): ~10–200 events per plan and ≤20 properties per event. Fewer than ~10 makes funnel analysis impossible; more than ~200 becomes unnavigable. For us, **stay near the low end (~12–15)**. [7]

Avoiding event sprawl — the SMB rule. Track **outcomes tied to business goals**, not every interaction. “Tracking every interaction inflates data volume and obscures important patterns.” [6] Let **autocapture** (PostHog/others) handle low-value clicks/pageviews for free, and **hand-instrument only the lifecycle events** in §6. [8]

Governance = a one-page tracking plan + an owner per event. The taxonomy fails not from a bad spreadsheet but from **no enforcement gate**. The highest-cost decay modes throw *no error* (silent identity fragmentation, orphaned events whose author left) [7]. Minimum viable governance for a small team:

- A living **tracking plan** (table in §6) — every event has: name, trigger, key properties, **named owner**, client/server.
- A **PR checklist** for any tracking change: name passes convention · properties typed · owner field set · deprecations get a migration plan, not a silent rename. [7]
- Use **version labels** (proposed / active / deprecated / removed) rather than deleting. [6][7]

3. Identity — anonymous → identified stitching (the silent killer)

Most value-defining events happen *before* we know who the user is (landing, signup-started). The job of identity stitching is to **merge that anonymous history into the known store-owner profile** the moment they identify [9][10].

The model. Each browser gets an **anonymous/device id** (PostHog `distinct_id` , Segment `anonymousId`). On signup/login you call `identify(user_id)` , which binds the anon id to the real **user id** and **retroactively merges** the prior anonymous events into that profile [9][10][7].

The two-moment rule (do both or graphs fracture silently): [7]

1. Initialize identity **on first load** (bind the anonymous session).
2. Call `identify()` **after login/signup** (merge anon → known).

Skip either and you get the **highest-cost, error-free failure**: cohorts that are silently wrong because one human is split across many ids, or many humans collapsed into one shared device. [7]

\$set vs \$set_once . Use `$set` for mutable attributes (current plan, store vertical) and `$set_once` for **immutable acquisition context** (first-touch channel, referral source) — overwriting acquisition source

destroys attribution permanently. [7]

Multi-tenant caveat for Print-Flow-360. We have two identity grains: the **store/tenant** and the **individual user**. Pick the store (tenant uuid) as the primary analytics identity for lifecycle/North-Star analysis (the North Star is *stores* with an order in 7 days), and carry `user_id` as a property — never expose raw tenant uuids in any user-facing analytics label (house rule §0). Merge-lag can create ~24h discrepancies in fresh cohorts — don't panic-diagnose a day-old funnel [1].

4. Server-side vs client-side — fire revenue events on the server

The principle: client-side events travel through the user's browser, where **ad-blockers, Intelligent Tracking Prevention, and JS failures drop a chunk of them**; server-side events go straight from our backend and **cannot be blocked** [11][12][13]. The events most likely to be lost client-side are exactly the **high-value ones** (signups, orders, payments) because privacy-conscious/blocker users skew toward exactly those moments [11].

Data-loss reality (confidence-flagged):

Claim	Value	Confidence
Ad-blocker <i>adoption</i> (share of internet users)	~29.5% global, ~33% US (Q2 2025)	Solid [14]
Ad-blocker <i>analytics-data-loss</i> (events dropped client-side)	~20–40% “in some industries”	Directional / vendor-stated, unaudited [11][13][7]
“One fintech lost 40% of conversion data”	single anecdote	Directional [11]

The hybrid model (recommended): [7][12]

- **Client-side (autocapture)** for low-stakes behavioral data: pageviews, clicks, in-app navigation. Cheap, no eng cost, blocker-loss is acceptable here.
- **Server-side** for the **money/lifecycle truth**: `order_received` , `payment_succeeded` , `subscription_started/changed/canceled` , `store_published` , `first_order_received` . These already happen in our Laravel backend (orders, payments, tenancy) — fire the analytics event in the **service layer**, not the Vue component.
- For an event you genuinely need on both sides, give the two **different names** to avoid double-counting [7].

Why this fits Print-Flow-360 specifically: our order/payment/publish logic *already lives server-side* (`app/Services/...` , per `CLAUDE.md` layering). Emitting the revenue events from there is both more reliable **and** less work than re-deriving them from blockable client events. This is the single highest-ROI instrumentation decision in the doc.

5. Tooling trade-offs for a cost- and eng-constrained team

Tool	Free tier (2026, re-verify)	Model	Strengths	Watch-outs	Confidence
PostHog	~1M events/mo + 5k session replays + feature flags, unlimited seats	Per-event (~\$0.00031/event after free)	Bundles analytics + replay + flags + A/B + surveys; self-hostable ; autocapture; EU data option	Per-event cost scales with volume; busier UI	Solid [15] [16]
Mixpanel	Up to ~1M MTU / generous event tier (free)	Per monthly-tracked-user (MTU)	Fast, PM-friendly funnels/retention	MTU pricing can surprise; less bundled	Solid [15] [16]
Amplitude	Starter up to ~10M events/mo (core charts)	Event/MTU tiers	Best-in-class behavioral analytics, governance/codegen	Enterprise-priced beyond starter	Solid [15] [16]
GA4	Free	—	Free, ubiquitous	Sampling, hard limits (40-char names, 25 params/event), session-not-user model, weak product funnels	Solid [7]
Self-host via Postgres	Infra cost only	Roll-your-own event table	Full control, no vendor; we already run Postgres	Build/maintain cost is the trap — no funnels UI, no replay, you build governance yourself	Directional

Recommendation for us: PostHog. It maximizes signal-per-dollar-and-per-eng-hour for a small team — autocapture removes most manual instrumentation, the free tier comfortably covers our SMB volume, and bundled session replay is (per PostHog) *“the most useful analytical tool for early-stage startups”* for turning a “where they dropped” into a “why” [8]. Self-host option aligns with our multi-tenant/EU-data posture. **Do NOT build a Postgres-only analytics layer** — we’d reinvent funnels, retention, replay, and governance with eng time we don’t have. (A lightweight Postgres `events` table is fine only as a server-side *source of truth* feeding PostHog, not as the analytics product itself.)

6. DELIVERABLE — Print-Flow-360 starter tracking plan

A deliberately **small (~14-event) plan** mapping the real lifecycle. One convention: `object_action`, snake_case, past tense. **S = server-side (reliable)**, **C = client-side (autocapture-OK)**. Every event has an owner. This is the governance doc — keep it living; add an event only when an existing one can't answer a real question.

Event name	Trigger (when it fires)	Key properties	Owner	Side
<code>landing_viewed</code>	Marketing/landing page load	<code>source</code> , <code>utm_*</code> , <code>device</code>	Marketing	C
<code>signup_started</code>	Owner opens signup form	<code>source</code>	Growth	C
<code>signup_completed</code>	Account/tenant created	<code>store_vertical</code> (from “what do you sell?”), <code>source</code>	Growth	S
<code>onboarding_question_answered</code>	“What do you sell?” answered	<code>store_vertical</code>	Product	C
<code>product_added</code>	First/any product saved	<code>product_type</code> , <code>is_first_product</code> , <code>price_set</code> (bool), <code>source</code> (<code>demo_seed</code> / <code>manual</code>)	Product	S
<code>price_set</code>	Owner sets/changes a product price	<code>product_type</code> , <code>used_default</code> (bool)	Product	S
<code>store_published</code>	Store goes live	<code>products_count</code> , <code>time_since_signup_hours</code>	Product	S
<code>store_link_shared</code>	Owner shares/copies storefront link	<code>channel</code>	Growth	C
<code>order_received</code>	Any order placed on a storefront	<code>order_total</code> , <code>currency</code> , <code>is_first_order</code> , <code>customer_type</code> (guest/b2b)	Orders	S
<code>first_order_received</code>	The store's first-ever order (North Star)	<code>order_total</code> , <code>currency</code> , <code>time_since_publish_hours</code>	Orders	S
<code>reorder_received</code>	Returning customer reorders	<code>order_total</code> , <code>days_since_last_order</code>	Orders	S
<code>studio_used</code>	Owner/customer uses design studio	<code>is_template</code> , <code>personalize_mode</code> (bool)	Product	C
<code>card_added</code>	Payment method captured (the upgrade ask)	<code>trigger</code> (<code>post_first_order</code> / <code>calendar</code> / <code>manual</code>)	Billing	S
<code>subscription_changed</code>	Trial → paid, plan change, or cancel	<code>from_plan</code> , <code>to_plan</code> , <code>mrr_delta</code> , <code>reason</code> (on cancel)	Billing	S

User properties to **\$set** : `plan`, `store_vertical`, `is_b2b`, `store_status` (trial/live/downgraded).

\$set_once : `first_touch_source`, `signup_date`.

The two headline funnels to build from this plan:

1. **Activation (North Star), strict-ish, 7–14d window:** `signup_completed` → `store_published` → `first_order_received`. Segment by `store_vertical` and `source`. This is the *one* funnel to review weekly.
2. **Storefront purchase, strict, short window:** `landing_viewed` (storefront) → `product_added` (cart) → checkout → `order_received`. Strict because checkout is linear.

What to A/B test (on our funnel, never on borrowed lift %): conversion-window length for activation; behavioral (`post_first_order`) vs. calendar `card_added` trigger; demo-seed vs. blank-start effect on `product_added` → `store_published`.

What to deliberately NOT instrument now: per-button clicks, scroll depth, hover/heatmap events, micro-interactions inside the studio, per-field form analytics. Autocapture + session replay cover these for free; hand-instrumenting them is the sprawl trap.

7. What this means for Print-Flow-360

Instrument (do now):

1. **Fire all 7 server-side events** (`signup_completed` , `product_added` , `price_set` , `store_published` , `order_received` / `first_order_received` / `reorder_received` , `card_added` , `subscription_changed`) **from the Laravel service layer** — they already run there; this is reliable *and* less work.
2. **Adopt PostHog**, turn on autocapture (covers all the client-side low-value events for free), and hand-instrument only the ~14 plan events.
3. **Wire identity correctly:** init anon id on first load; call `identify(tenant_uuid)` at signup/login; `$set_once` the first-touch source. This prevents the silent-cohort bug.
4. **Build the activation funnel** (`signup_completed` → `store_published` → `first_order_received`) strict-ordered, 7–14 day window, segmented by vertical and channel. Review the worst-converting step weekly.
5. **Make the tracking plan (\$6) a checked-in doc with a PR checklist** — owner field required, names are fixed strings, deprecations get migration plans.

Ignore (resist the temptation):

- Don't build a Postgres-only analytics product — reinventing funnels/retention/replay wastes the eng time we don't have.
- Don't optimize the storefront micro-funnel before the **activation** funnel is measured and healthy (PostHog: premature funnel optimization “hides bigger problems” [8]).
- Don't track every click manually — autocapture exists; sprawl is the enemy.
- Don't trust borrowed lift/data-loss percentages as targets.

A/B test (on our own funnel): activation conversion-window length · `post_first_order` vs calendar `card_added` trigger · demo-seed vs blank-start on the product → publish step.

Connects to prior research: this operationalizes `CONVERSION_FUNNEL_RESEARCH_2026-06-15.md` §7-8 (the North Star stores with an order in 7 days is the `first_order_received` event; the behavioral upgrade trigger is the `card_added` `trigger: post_first_order` property).

8. Folklore warnings — stats commonly mis-attributed or fabricated

- **“Ad-blockers cause 25–40% analytics data loss.”** Ad-blocker *adoption* (~29–33% of users) is well-measured [14]; the *event-loss* figure is **vendor-stated and explicitly unaudited** [7][11][13]. Use it to justify server-side capture (directionally true), not as a precise target.
 - **“You lose a \$5,000 sale every time a pixel is blocked.”** Vendor illustration, not data [11].
 - **“One fintech lost exactly 40% of conversions to ad-blockers.”** Single uncited anecdote [11]. Directional at best.
 - **Amplitude’s “10–200 events / ≤20 properties”** is *functional guidance*, not a hard platform limit — don’t cite it as a rule [7].
 - **GA4 hard limits ARE real and load-bearing** (40-char event names, 25 params/event, name must start with a letter) — these are platform constraints, not folklore [7]. If we ever dual-ship to GA4, names must fit.
 - **“Object-Action framework” attribution.** It’s a shared industry convention documented by Amplitude, Mixpanel, and PostHog — fine to cite, but it has no single canonical inventor; don’t attribute it to one vendor as if proprietary [6][7].
 - **Tool free-tier numbers drift.** The PostHog/Mixpanel/Amplitude free-tier figures here are 2026 snapshots [15][16] — re-verify before committing; vendors change them frequently.
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Print-Flow-360 Metrics & Economics: Index and Operating System

Source: readme/research/metrics-analytics/METRICS_AND_ECONOMICS_INDEX_2026-06-16.md

IN SHORT — KEY TAKEAWAYS

This document is a guide to which business numbers Print-Flow-360 should watch — and which ones to ignore. It covers five research topics at once: how much it costs to get a customer, whether customers come back to reorder, how monthly revenue moves, how to track the steps from signup to first order, and how to find out where new customers come from. The main message is simple: track a small, honest set of numbers; ignore the complicated ones that big companies use.

The main parts explained simply:

- **North Star goal** — The one number everything else is built around: did a new store go live AND place its first order within 7 days of signing up? Every other metric is judged by whether it helps reach this goal.
- **CAC Payback (how long to earn back what you spent to get a customer)** — Instead of complex formulas, track one thing: how many months of subscription fees does it take to recover the full cost of winning a new customer? Include founder time and free tool costs. Keep this number as low as possible.
- **Cohort retention triangle (do customers come back to reorder?)** — Group customers by the month they signed up and track whether they placed repeat print orders. This shows clearly whether the business keeps customers or slowly loses them. Build it with plain database queries — no expensive software needed.
- **MRR / ARR / NRR (monthly and yearly subscription money)** — Count only the fixed subscription fee as recurring revenue. Keep order volume (GMV) on a separate line so the numbers stay clean and honest. For a small-business SaaS, keeping about 97% of revenue year-over-year is healthy — the 120% target you read about online is for large enterprise software, not us.
- **Funnel tracking (the steps from signup to first order)** — Set up roughly 14 key events that mark a store's journey from “just signed up” to “first order received.” Fire these from the server, not the browser, so no events are missed.
- **Attribution (where do new customers come from?)** — Skip complicated tracking tools. Ask one plain question at signup: “How did you hear about us?” Also capture the first link a visitor clicked. That is enough to make good decisions.
- **Folklore warnings (popular “facts” that are actually wrong)** — Several widely-shared SaaS statistics are exaggerated or made up. This document flags the worst ones so the team does not set the wrong targets based on bad numbers.

What to do with this: Start by building the signup-to-first-order funnel and the “How did you hear about us?” question using your existing database — no new tools needed. Add the monthly cohort retention view next. Only buy analytics software if those plain database reports stop being enough.

Internal index/synthesis doc. Researched 2026-06-16 via a multi-agent workflow (five parallel research streams — SaaS unit economics, cohort retention, funnel/event taxonomy, MRR/ARR/NRR, and low-ACV SMB attribution — each independently fact-checked, then synthesized here). Benchmarks in the five companion docs are flagged by confidence; treat vendor-blog percentages as **hypotheses to A/B-test on our own data**, not guarantees. Several widely-repeated SaaS-metrics “folklore” numbers were found to be mis-attributed or fabricated — each companion doc carries an explicit folklore-warnings section, and the worst offenders are summarized below.

Executive Summary

Five companion research docs were written for the Print-Flow-360 team covering the full span of SaaS metrics and economics. Read together, they argue for **one coherent operating system**, not five separate dashboards. The unifying thesis: Print-Flow-360 is a **low-ACV, self-serve SMB print SaaS** sold to non-technical print-shop owners, so almost every “standard” SaaS metric you read about on vendor blogs is calibrated for the wrong company — enterprise-ACV, sales-led, \$15M+ ARR public-company benchmarks that will mislead a cash-constrained team if adopted as targets.

The opinionated headline, consistent across all five streams: **at this stage, track a deliberately tiny set of metrics, built almost entirely in plain Postgres SQL on data you already have, and tie every one of them to the existing North Star — “store live + first order in 7 days.”** Specifically: make **margin-adjusted CAC Payback (in months)** your single headline economic metric; make a **store-signup-month retention cohort triangle** your single headline retention artifact; keep **one clean MRR/ARR base** (subscription fees only, GMV reported separately); measure acquisition channels by **activated stores via a self-reported “How did you hear about us?” survey + first-touch UTM** (skip multi-touch attribution entirely); and instrument a **~14-event server-side funnel** whose only job is to measure activation. Everything else — Rule of 40, Rule of X, Magic Number, NRR-chasing toward enterprise 120%, multi-touch/DDA/MMM attribution, ChartMogul/Amplitude/Mixpanel — is **defer until later**. The cheapest tooling path (Postgres + a survey field + optionally PostHog) carries you a long way before any paid analytics platform earns its keep.

Companion docs — table of contents

#	Doc	Headline recommendation
1	SaaS Unit Economics (CAC, LTV, Payback, Rule of 40)	Make margin-adjusted CAC Payback (months) the single headline economic metric on a fully-loaded CAC (founder time + free design tool serving cost), read alongside cohort churn; ignore Rule of 40 / Rule of X / Magic Number until repeatable measured S&M spend at ~\$5M+ ARR.
2	Cohort Analysis & Retention Curves	Kill the single blended churn %; make a store-signup-month retention triangle in plain Postgres SQL, with “retained” defined on the print reorder cadence (not logins), the headline retention artifact — don’t buy ChartMogul/Amplitude/Mixpanel until that view is genuinely the bottleneck.
3	Funnel Analytics & Event Taxonomy	Adopt PostHog , fire all money/lifecycle events server-side from the Laravel service layer , identify on the tenant uuid , and ship a deliberately small ~14-event plan whose single job is to measure <code>signup_completed → store_published → first_order_received</code> within 7–14 days.
4	MRR / ARR / NRR Mechanics	Keep ONE clean recurring-revenue base : MRR/ARR = subscription plan fees only (normalized monthly); report platform GMV and any take-rate revenue on a separate line ; accept ~97% NRR as healthy for low-ACV SMB rather than chasing the wrong 120% enterprise target.
5	Attribution for Low-ACV SMB	Skip multi-touch attribution entirely : ship a plain-language “ How did you hear about us? ” field + first-touch UTM capture, report channels by activated stores (North Star), and run a holdout test only when about to scale spend on a channel.

Recommended metrics stack for Print-Flow-360

The point of this index is restraint. A small team serving non-technical SMB owners should resist the temptation to build the dashboard a Series-B company has. Below is the short list that actually earns its keep at this stage.

The single North Star (reuse, do not reinvent)

Store live + first order within 7 days of signup. This is already established in

`CONVERSION_FUNNEL_RESEARCH_2026-06-15.md` as the activation event and `ACQUISITION_CHANNELS_2026-06-15.md` as the channel-quality yardstick. Every metric below should be read *through* this event — channels are judged by activated stores, cohorts are keyed on signup month, the funnel terminates on `first_order_received`, and CAC payback is computed against revenue from activated stores.

Track these — weekly

Metric	Source doc	Why now
New signups → stores published → first order (the activation funnel, by cohort week)	[3]	The North Star, instrumented; your single biggest leak lives here
Activation rate (% of signups hitting North Star within 7 days)	[3]	Leading indicator of everything downstream
Net new MRR + the 5-part movement (new / expansion / contraction / churn / reactivation)	[4]	One clean recurring-revenue base; catch churn early
Self-reported “How did you hear about us?” tally, by activated stores	[5]	Cheapest honest channel signal; no MTA needed

Track these — monthly

Metric	Source doc	Why now
Margin-adjusted CAC Payback (months), fully-loaded CAC	[1]	The headline economic metric for a cash-constrained team
Store-signup-month retention cohort triangle (reorder-cadence “retained”)	[2]	Replaces the blended churn %; shows whether the bucket leaks
GRR and NRR (SMB-calibrated: ~90% GRR, ~97% NRR are healthy)	[2][4]	Health of the base; do NOT target enterprise 120%
GMV and any take-rate revenue (separate line, never blended into MRR)	[4]	Keeps the recurring-revenue base clean and honest

Ignore until later (explicitly)

- **Rule of 40, Rule of X, Magic Number** — meaningful only with repeatable measured S&M spend at ~\$5M+ ARR (most reliable \$15–20M+/public). [1]
- **NRR-chasing toward 120%** — that’s an enterprise/expansion-led target; ~97% is the right SMB number. [4]
- **Multi-touch attribution, data-driven attribution (DDA), media mix modeling (MMM)** — all overkill for a few-channel self-serve motion. [5]
- **Paid analytics platforms** (ChartMogul, Amplitude, Mixpanel) — don’t buy until a Postgres view is genuinely the bottleneck, not before. [2][3]
- **N-day / fixed-window retention curves** — use unbounded/bracketed retention on the reorder cadence instead. [2]

The cheapest tooling path

1. **Postgres SQL on data you already have** — the cohort triangle, MRR waterfall, GRR/NRR, and CAC payback are all plain queries against existing tables. Start with one **materialized cohort view + a simple admin heatmap**. This is ~90% of the value at ~0% of the platform cost. [2][4]
 2. **A self-reported attribution field** — a single “How did you hear about us?” question at signup, plus first-touch UTM capture. No attribution vendor required. [5]
 3. **PostHog (only when you need event-level funnel granularity)** — server-side events from the Laravel service layer, identified on tenant uuid, ~14-event plan. Defer even this until the SQL funnel can’t answer your activation questions. [3]
 4. **Defer heavy analytics platforms indefinitely** — revisit only when a specific question can’t be answered cheaply.
-

Sequencing — what to instrument first → last

The companion docs agree on order. Do not parallelize; each step makes the next cheaper and tells you whether the next is even needed.

1. **First — the activation funnel in SQL + the self-reported attribution field.** With one query you get the North Star, your biggest leak, and honest channel signal. Add the “How did you hear about us?” field and first-touch UTM at signup. (Docs [3], [5].)
 2. **Second — the store-signup-month cohort retention triangle (materialized Postgres view + admin heatmap).** Define “retained” on the print reorder cadence. This replaces the blended churn % and reveals whether the bucket leaks. (Doc [2].)
 3. **Third — the MRR movement waterfall + GRR/NRR, one clean base.** Subscription fees only; GMV/take-rate on a separate line. (Doc [4].)
 4. **Fourth — margin-adjusted CAC Payback on a fully-loaded CAC.** Requires the activation/cohort data from steps 1–2 to attribute revenue to activated stores; include founder time and the free design tool’s serving cost. (Doc [1].)
 5. **Fifth (only if needed) — PostHog and event-level funnel granularity.** Adopt when the SQL funnel can no longer answer activation questions. (Doc [3].)
 6. **Last / maybe never — Rule of 40 / Rule of X / Magic Number, multi-touch attribution, paid analytics platforms.** Revisit at ~\$5M+ ARR or when scaling spend on a channel (then run a holdout test, not an MTA build). (Docs [1], [5].)
-

Folklore warnings (summary — see each doc for the full list)

Adopting these as fact is a common and expensive mistake; each is debunked or re-sourced in its companion doc.

- The “**5% retention increase → up to 95% profit**” stat is **Bain/Reichheld** (a 25–95% range from 1990s studies), not original HBR research — attribute carefully. [1][2]
- **NRR ~120% is “healthy”** is an **enterprise/expansion-led** target, not an SMB one — ~97% is right for us; chasing 120% optimizes the wrong thing. [4]
- “**>5:1 LTV:CAC means you’re under-investing**” is not actually supported by The SaaS CFO; treat as a soft caveat, not a rule. [1]
- The **Mixpanel “2-second-to-90-day attribution window”** figure is fabricated — the real mechanics are a 2-second grace period and a separate (default ~7-day, up to 366-day) window. [3]
- **Rule of 40 has a firm floor of \$X ARR** — it doesn’t; it’s a range (~\$5M+ minimum, most reliable at \$15–20M+/public). [1]
- **Salesforce / “90% use last-touch”** — replaced with eMarketer’s ~78% last-click; the W-shaped ACV/cycle numbers commonly cited are fabricated. [5]
- Optifai / vendor-cohort CAC-payback and various **First Page Sage** 1–7-month figures are **self-serve floors or undisclosed-methodology vendor cohorts**, not targets. [1]

Sources

These are the five companion research docs, each with its own fully-cited Sources section. This index does not introduce new external citations; it synthesizes theirs.

1. **SaaS Unit Economics (CAC, LTV, Payback, Rule of 40)** — [readme/METRICS_SAAS_UNIT_ECONOMICS_2026-06-16.md](#)
2. **Cohort Analysis & Retention Curves** — [readme/METRICS_COHORT_RETENTION_2026-06-16.md](#)
3. **Funnel Analytics & Event Taxonomy** — [readme/METRICS_FUNNEL_EVENT_TAXONOMY_2026-06-16.md](#)
4. **MRR / ARR / NRR Mechanics** — [readme/METRICS_MRR_ARR_NRR_2026-06-16.md](#)
5. **Attribution for Low-ACV SMB** — [readme/METRICS_ATTRIBUTION_LOW_ACV_SMB_2026-06-16.md](#)

Related prior research (for grounding; do not contradict):

- [readme/CONVERSION_FUNNEL_RESEARCH_2026-06-15.md](#) — North Star = store live + first order in 7 days; no-card 14-day reverse trial.
- [readme/ACQUISITION_CHANNELS_2026-06-15.md](#) — PRIMARY founder-led outreach + print communities; SECONDARY BOFU SEO + free design tool; SKIP paid search/affiliates.

Attribution Models for Low-ACV SMB (Why Multi-Touch Is Often Overkill)

Source: readme/research/metrics-analytics/METRICS_ATTRIBUTION_LOW_ACV_SMB_2026-06-16.md

IN SHORT — KEY TAKEAWAYS

This document is about figuring out how new customers find your business — and whether you really need expensive, complicated tracking tools to do it. The short answer is: no, you don't. For a small print-software business like this one, a simple “How did you hear about us?” question on the signup form does the job better than the fancy systems that big companies pay for.

The main parts explained simply:

- **The tracking-method menu** — There are many ways to track where customers come from, ranging from simple (“credit the first place they found us”) to very complex (“split credit across every step of the journey using a machine-learning model”). This document lists all of them and explains which ones are actually worth using for a small business.
- **Why the complicated methods are overkill for us** — The big tracking systems work only when you have hundreds of new customers every month. If your numbers are smaller, those systems just show you confident-looking guesses, not real answers. They also can't see customers who heard about you from a friend, a Facebook group, or a community forum — which is exactly how most of our best customers find us.
- **Self-reported attribution (the survey field)** — This just means asking customers directly, “How did you hear about us?” with a short dropdown list. It costs almost nothing to build, and it captures word-of-mouth and community referrals that no tracking software can ever see. About half of people give a useful answer.
- **UTM tracking links** — When you share a link (on social media, in an email, anywhere), you can add a small label to the end of the URL. When someone clicks it and signs up, you know where they came from. Store that label once at sign-up and you have reliable first-party data that doesn't depend on cookies or third-party trackers.
- **Incrementality testing (the one experiment worth running)** — Instead of tracking every click, pause one channel (say, stop posting in a particular forum) for a few weeks and see if sign-ups drop. That tells you whether that channel is actually bringing in customers or just taking credit. This is the only method that proves cause-and-effect.
- **Using both methods together** — When the survey says “a friend told me” but the tracking says “Google,” trust the survey. The tracking misses the real reason; the survey surfaces it.

What to do with this: Add a “How did you hear about us?” dropdown to the sign-up form right away — it is the single highest-return action described here. At the same time, tag every shared link with a UTM label and save that label when someone signs up. Skip every other tracking tool until the business is much larger.

Researched 2026-06-16 via a multi-agent workflow (research → adversarial fact-check → synthesis). This doc is grounded in vendor and analyst material from 2024–2026. **Treat every vendor-blog percentage as a hypothesis to A/B-test against our own data, not a guarantee.** Several widely-repeated “SaaS-metrics folklore” numbers are explicitly flagged below — do not launder them into our planning decks as hard benchmarks. Where a stat survived independent verification it is marked Verified/Solid; where it traces to a single vendor self-test or a mutually-cited blog chain it is marked Directional or Fabricated-avoid.

Executive Summary

Headline recommendation: Skip multi-touch attribution entirely. Ship a plain-language “How did you hear about us?” field + first-touch UTM capture, report channels by *activated* stores (North Star), and run a holdout test only when you’re about to scale spend on a channel.

Print-Flow-360 is a low-ACV, self-serve print SaaS sold to non-technical print-shop owners, with a deliberately concentrated acquisition strategy (founder-led outreach + print communities as PRIMARY; BOFU SEO + free design tool as SECONDARY — see [readme/ACQUISITION_CHANNELS_2026-06-15.md](#)). For that shape of business, the multi-touch attribution (MTA) industry is selling us a solution to a problem we don’t have:

- **Our sales cycle is days, not months, with few touchpoints.** There is almost nothing for a multi-touch model to split credit across. Single-touch (first/last) is genuinely sufficient.
- **We don’t have the conversion volume** to make data-driven/algorithmic attribution anything but a noise-fitter (it wants ~600–1,000 conversions/month to be stable [7][8]).
- **Cookies are gone.** Roughly **40–60% of conversions are now invisible to cookie-based tracking** [1][3], so an MTA platform would give us a confident-looking but half-blind picture — and our highest-priority channels (word-of-mouth in print communities, the free design tool) are exactly the “dark social” that tracking can’t see at all.
- **We have ~2 engineers.** The maintenance tax of an attribution model would exceed its decision value.

The single highest-ROI attribution upgrade for a team like ours is **self-reported attribution (SRA)** — one survey field on a high-intent form — corroborated by clean first-party UTMs. This mirrors the strongest, most-cited industry position (Refine Labs / Chris Walker, adopted across 100+ B2B companies [4]) and aligns one-for-one with the channels our acquisition doc told us to bet on. Build the lightweight stack; ignore MTA, DDA, and MMM.

1. The attribution-model zoo (and where it stops being worth it)

Attribution assigns conversion credit to marketing touchpoints. The models form a spectrum from trivial to heavyweight:

Model	Rule	Best for	Fit for Print-Flow-360
First-touch	100% credit to the discovery channel	Seeing what <i>drives demand</i>	✓ Yes — pairs with SRA + signup UTM
Last-touch	100% credit to the final channel before conversion	Seeing the <i>closing</i> channel	✓ Yes — cheap default; pair with first-touch
Linear	Equal credit to every touch	Simple multi-touch	✗ Flattens influence; little to model in short cycles
Time-decay	More credit the closer a touch is to conversion	Longer considered purchases	✗ Overkill for fast self-serve
U-shaped / position-based	40% first + 40% last + 20% across the middle	Rules-based MTA, 3–6 mo cycles	✗ Adds nothing over first/last for us
W-shaped	30% first + 30% lead-creation + 30% opp-creation + 10% middle	Multi-stage <i>sales-led</i> motions	✗ We have no lead/opp stages
Data-driven (DDA)	ML/Shapley-style credit from observed paths	High-volume conversion data	✗ Below volume threshold it models noise

The first three rows (first/last/linear) are single-touch or trivial. U-shaped, W-shaped, and DDA are the heavyweight “**multi-touch attribution**” (MTA) end — and that’s the end that is overkill for us.

Single-touch formulas (the only two we’ll actually use)

```
First-touch: Credit(channel) = 1.0 if channel == first touchpoint, else 0
Last-touch:  Credit(channel) = 1.0 if channel == last touchpoint before conversion, else 0
```

First-touch maps cleanly to a self-reported survey and a UTM captured at signup — it answers “*what made them discover us?*” Last-touch is the de-facto default in nearly every analytics tool and answers “*what closed them?*” Use both; rely on neither alone.

The multi-touch formulas (for reference — we are not building these)

```
Linear:      Credit(each touch) = 1 / N      (N = number of touchpoints)
Time-decay:  Credit(touch_i) ∝ 2^(-Δt_i / halflife), normalized so Σ credit = 1
U-shaped:    40% first + 40% last + 20% split across middle touches
W-shaped:    30% first + 30% lead-creation + 30% opp-creation + 10% across the rest
DDA:         Algorithmic / Shapley-value credit from converting vs non-converting paths
```

2. MTA vs MMM vs Incrementality — three different jobs, not three brands of the same thing

A common mistake is treating these as competing products. They answer different questions:

Method	What it is	Proves causation?	Data appetite	Verdict for us
MTA (multi-touch attribution)	User-journey, click-path credit splitting	✗ Correlational	High conversion volume; privacy-fragile	Least valuable job for us
MMM (marketing-mix modeling)	Top-down regression of aggregate spend vs revenue	✗ Correlational	~2+ years of weekly spend/revenue data	Too data-hungry — skip
Incrementality / holdout	Controlled experiment: withhold a channel, measure lift	✓ Yes — the only causal method	A control group + a few weeks	Highest-signal test we can run

Marketers themselves rate MMM as more reliable than MTA — in eMarketer’s survey, **MMM was rated most reliable by ~28% vs MTA’s ~19%** [6]. But MMM needs years of weekly data we don’t have, and MTA can’t prove causation. That leaves **incrementality** as the method that actually earns its keep for an early SMB: a cheap, periodic holdout that answers the only question that matters before scaling spend — “*would this conversion have happened anyway?*”

Incrementality formula

```
Incremental lift %      = (ConvRate_exposed - ConvRate_control) / ConvRate_control
Incremental conversions = (ConvRate_exposed - ConvRate_control) × exposed_population
```

Run a randomized holdout (or geo-split) for at least a few weeks with a properly sized control group, then compare. Adoption is mainstream and growing: **~52% of US marketers use incrementality testing and ~36% plan to invest more in it over the next year** [6].

3. Why heavy MTA is overkill (and often misleading) for low-ACV SMB

Four structural reasons, all of which apply to Print-Flow-360:

1. **Short, cheap sales cycles.** A self-serve print-shop owner has a handful of touchpoints over days or weeks. There is little journey to model, so credit-splitting adds complexity without insight. Single-touch is *genuinely* sufficient — not a compromise.
2. **Low data volume per channel.** Data-driven/algorithmic models need hundreds of conversions *per conversion type* to be statistically reliable (see §4). An early SMB SaaS won’t hit that, so the “model” fits noise and hands you confident garbage.
3. **Privacy / cookie decay.** Third-party cookies and mobile IDs are largely gone. Cookie consent/match rates run roughly **40–60%**, and cookie-based tracking now **misses an estimated 30–50% of conversions** [1][3]. MTA is blind to a large share of journeys — and *especially* blind to the dark-social channels (community word-of-mouth, a friend’s recommendation, the free design tool) we are deliberately leaning on.

4. **Maintenance tax.** A 2-person team spends more time explaining and cleaning the attribution model than acting on it. *If you can't maintain it, it won't help you decide.*

Folklore flag — “you’re flying blind without multi-touch attribution.” This is vendor sales pressure, not analysis. The documented reality is the opposite for our segment: MTA is over-engineered below roughly \$5–20M revenue / fewer than ~5 active channels (these thresholds are reasonable rules of thumb, not sourced laws), and a simple SRA + UTM stack outperforms it on both cost and on capturing dark social.

4. Data-driven attribution: the volume reality (and the GA4 trap)

DDA is the most seductive model — “let the machine figure it out.” For us it’s a trap, for two reasons.

Metric	Value	Confidence	Source
Legacy Google UA threshold to enable DDA	600 conversions / 30 days (Search); + 15,000 Google Ads clicks in the Ads variant	Verified	[7]
Legacy Google UA threshold to maintain DDA	~400 conversions per conversion action + ≥ 2 path interactions over the trailing window	Verified	[7]
Practical DDA reliability floor (incl. GA4)	~600–1,000 conversions/month for stable results	Directional	[8]

Folklore flag — “you need 600 conversions to use data-driven attribution.” This is from **sunset legacy Universal Analytics** [7]. **GA4 removed the hard threshold and made DDA the default model.** So “600 conversions” is *outdated as a hard gate*. The honest framing: GA4 will *technically let you run* DDA at any volume, but below roughly several hundred conversions/month it is unreliable — and critically, **GA4 silently falls back to a rule-based/last-click model below threshold and does not notify you** [8]. You’d think you’re running DDA when you’re not.

Net: at our volume, DDA is either noise (if it ran) or secretly last-click (if it fell back). Either way, it’s last-click with extra steps. Just use last-click on purpose.

5. Self-reported attribution (SRA) — the SMB-grade workaround

A single survey field at signup/checkout: “*How did you hear about us?*” It is the highest-leverage attribution move for a team our size because it captures **dark social** — word-of-mouth, community/forum chatter, podcasts, events, a friend’s recommendation — that no tracking can see.

Its honesty cuts both ways: it surfaces the *most memorable/meaningful* touch (often the real demand driver), but it’s recall-biased, sometimes vague (“Google”), and only reflects the one person filling the form.

Best practice:

- Place it on a **high-intent form** (signup/checkout), not a low-intent newsletter box.
- Use a **curated dropdown of 5–8 plain-language options** + an “Other” free-text.
- Treat it as **one corroborating signal**, not gospel.

Metric	Value	Confidence	Source
SRA usable-response rate (single n=100 vendor self-test)	~70 of 100 filled the field; 49 of 100 gave actionable answers (~49%)	Solid (but n=100, one vendor)	[2]
Dark social is large (clean anchor)	B2B buyers complete 70–80% of research before contacting sales — much of it untrackable	Directional	[9]
Hybrid (tracked + self-reported) lead-source accuracy	One vendor reported ~81% — single-source, unverifiable	Fabricated-avoid	[5]

Folklore flag — the “podcasts drove 53% of revenue self-reported but 0% in software / 90% measurement gap” anecdote. A vivid, endlessly-repeated *single-vendor* story (Refine Labs). Use it only to *illustrate* that tracking misses dark social — the exact 53% / 0% / 90% figures are **not a generalizable benchmark**. Same caution applies to Refine Labs’ “97% of net-new ARR traced to dark social” and “99% of self-reported responses differ from last-touch”: directionally evocative, not laws. For the clean, citable version of “dark social is large,” use Forrester’s 70–80%-of-research-before-sales-contact figure [9] instead.

Folklore flag — “hybrid = 81% lead-source accuracy.” Could not be verified in any source; traces to one vendor study repeated across roundups. The *underlying point* (combining tracked + self-reported beats either alone) is sound; the precise **81% is not a benchmark — do not cite it as fact.**

6. UTM hygiene — the cheap, durable foundation

UTM parameters (`utm_source` , `utm_medium` , `utm_campaign`) tagged on every outbound link, **captured at signup and stored first-touch on the store/customer record**, give you reliable *first-party* source data that survives the cookie apocalypse — because *you* store it, not a third party. Combined with a consistent naming convention, UTMs + a single first-touch capture cover the *tracked* half of attribution at near-zero ongoing cost. This is the “simple first/last touch” layer that replaces an entire MTA platform for an SMB.

The known blind spot: UTMs only see *clickable* links. They will report “direct/organic” for the print-shop owner who heard about us in a Slack group and typed the URL. That gap is exactly what the SRA field fills (§5) — which is why we run both and triangulate (§8).

7. Where last-touch and MTA actually stand in the market

Useful as a reality check on vendor hype: the “advanced” models are far less used than the marketing implies, and the boring default still dominates.

Metric	Value	Confidence	Source
Last-click is the de-facto default	~78.4% of marketers use last-click attribution; only ~21.5% are confident it reflects long-term impact	Directional	[6]
Perceived reliability of methods	MMM ~28% rated most reliable > MTA ~19% > unified ~19%	Solid	[6]
MTA adoption at sub-\$5M revenue	~44% of sub-\$5M companies use MTA	Directional	[11]
W-shaped fit	Suits mid-market B2B with multi-stakeholder, multi-month cycles (commonly 6–18 mo); algorithmic/custom only for high-volume enterprise with enough conversion data	Directional	[12]

Folklore flag — “~90% of brands use last-touch (Salesforce 2022)” and “~67% last-touch adoption.” The Salesforce/90% figure could not be located; we’ve replaced it with the verifiable eMarketer 2024 figure (~78.4% use last-click, ~21.5% trust it) [6]. The “~67%” sub-figure is unverified vendor folklore — **dropped**. The *takeaway* is unchanged: last-touch is the de-facto default, **not** a best practice.

Interpretation note on the 44% stat. CaliberMind’s table shows 44% of sub-\$5M firms use MTA as *an adoption fact* [11]. But the same table shows enterprise (\$250M–\$1B) at 73% and a non-monotonic middle — so small firms do **not** lead adoption. The “many of them are over-engineering” read is **our analysis**, consistent with §3, not a CaliberMind conclusion.

Folklore flag — fabricated W-shaped specifics. Earlier drafts cited W-shaped as fitting “\$50K–\$150K ACV / 4–8 month cycles” and algorithmic for “\$250K+ / 50+ conversions over 6–12mo.” Those precise numbers are **not in the cited source**, which segments by *company ARR* (\$1M–\$10M → W-shaped; \$10M+ → algorithmic) [12], not deal ACV, and uses one illustrative example. The fabricated ranges have been **stripped**. Either way: W-shaped needs lead/opp stages we don’t have.

Folklore flag — MTA/MMM adoption stats (“75% use MTA,” “47% in 2026 up from 31%,” “MMM is now free/easy with Meridian/Robyn/PyMC”). These float around vendor blogs with inconsistent denominators and conflate rules-based MTA with full algorithmic platforms. Treat as *vibe*, not data. And while the MMM libraries *are* free, MMM still needs ~2+ years of weekly spend/revenue data to

mean anything — **citing it as an SMB option is a category error for an early-stage print SaaS.**
Skip.

8. Hybrid corroboration beats either method alone

Tracked attribution and self-reported attribution disagree *constantly*. Rather than trust one, **triangulate**:

- **Where first-touch UTM and self-reported agree** → high confidence. Bank it.
- **Where they diverge** (tracking says “direct/organic” but buyers say “a friend told me” or “your free design tool”) → the **self-reported answer usually reveals the real demand driver**, and the tracked one is just the last clickable step.

For Print-Flow-360 specifically, this is the whole ballgame: last-touch “direct” will routinely *hide* that referral and the free design tool are the actual engine — precisely the dark-social channels our acquisition strategy prioritized. The hybrid view is what stops us from defunding our best channel because a dashboard couldn’t see it.

What this means for Print-Flow-360

Concrete, scoped to a small non-technical customer base and ~2 engineers. **Instrument simple things well; do not model.**

BUILD (in priority order)

1. **“How did you hear about us?” field at signup.** Curated dropdown of 5–8 plain-language options + “Other” free-text. No jargon — never the words “attribution,” “channel,” or “source” in the UI. Suggested options tailored to our actual bets:
 - “A friend or another print shop told me”
 - “Google search”
 - “Facebook / Instagram”
 - “A print-industry group or forum”
 - “YouTube / a video”
 - “Saw your free design tool”
 - “Other” (free text)

This is the **highest-ROI attribution work** and directly measures the founder-led-outreach + print-communities + free-design-tool channels from `readme/ACQUISITION_CHANNELS_2026-06-15.md`.

2. **First-touch UTM capture.** Read `utm_source` / `utm_medium` / `utm_campaign` on the landing page and **persist them first-touch on the store/customer record at signup** in a small additive JSON column. Follow the existing `custom_field_values` pattern — **additive only, never repurpose an existing column** (per `CLAUDE.md` §5). This is the first-party tracked half that survives cookie decay.

3. **One internal “Where signups come from” report.** A weekly table joining the self-reported answer + first-touch UTM source, **segmented by whether the store hit the North Star (store live + first order in 7 days)** from `readme/CONVERSION_FUNNEL_RESEARCH_2026-06-15.md`. Reuse existing admin list/report patterns — **do not buy or build a dashboard product**. Report channels by **activated stores, not raw signups**, so a vanity channel that brings tire-kickers is visibly deprioritized.

INSTRUMENT, don’t model (engineering correctness)

- Store the **raw** self-reported answer and the **raw** first-touch UTM so they round-trip: **validate** → **save** → **read back** → **render**. Per `CLAUDE.md`’s silent-lie rule, a survey field collected in the UI but missing from the `FormRequest` `rules()` is **silently discarded** — the owner sees “Saved” but the data is gone.
- **Write a test** asserting both fields persist, mirroring `tests/Feature/Storefront/AddressAttentionToTest.php`. A field with no such test is not done.

A/B TEST / experiment (causal reads, periodic — not continuous)

- **Run a holdout, not a tracker.** When the founder is about to pour time or money into a channel (a specific print-community sponsorship, a paid experiment), run a lightweight **incrementality test**: pause or geo-split that channel for at least a few weeks and compare signup *and activation* rate against a control. This causal read is worth more than any attribution model for a base our size.
- **A/B the SRA field itself** if response quality is poor: dropdown order, option wording, optional vs required. Aim to beat the ~49% usable-answer baseline [2].

IGNORE (do not spend a single sprint on)

- ❌ A multi-touch attribution platform (MTA) — overkill for short, cheap, few-touch cycles.
- ❌ Data-driven / algorithmic attribution (DDA) — we lack the ~600–1,000 conversions/month for it to be anything but noise or silent last-click [7][8].
- ❌ Marketing-mix modeling (MMM) — needs ~2+ years of weekly data; category error at our stage.
- ❌ U/W-shaped models — they assume lead/opp sales stages we don’t have.

CORROBORATE divergences (the judgment call)

When tracked says “direct/organic” but buyers say “a friend told me” or “your free design tool,” **trust the self-reported demand driver** and credit word-of-mouth / the free tool. Don’t let last-touch “direct” bury the fact that referral and the free design tool are the real engine — those are exactly the channels the acquisition doc told us to double down on.

Sources

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- [13] HockeyStack — *Understanding Different Attribution Models and When to Use Them.* <https://www.hockeystack.com/blog-posts/different-attribution-models>
- [14] Digital Applied — *Marketing Mix Modeling 2026: MMM vs Attribution Playbook.* <https://www.digitalapplied.com/blog/marketing-mix-modeling-2026-mmm-vs-attribution-playbook>

Related internal docs

- [readme/CONVERSION_FUNNEL_RESEARCH_2026-06-15.md](#) — AARRR funnel, no-card 14-day reverse trial, North Star (store live + first order in 7d). The SRA report should segment by this North Star.
- [readme/ACQUISITION_CHANNELS_2026-06-15.md](#) — PRIMARY founder-led outreach + print communities; SECONDARY BOFU SEO + free design tool; SKIP paid search/affiliates. The SRA

dropdown options are designed to measure exactly these bets.

Cohort Analysis & Retention Curves (vs. Simple Churn %)

Source: [readme/research/metrics-analytics/METRICS_COHORT_RETENTION_2026-06-16.md](#)

IN SHORT — KEY TAKEAWAYS

This document is about how to measure whether your customers are sticking around — and why the simple “churn %” number most businesses track is misleading. It explains better ways to see if your store owners keep using Print-Flow-360 or quietly walk away, so you can fix problems before they quietly destroy the business.

The main parts explained simply:

- **Why the single churn number lies** — One overall “churn %” can look fine while things are quietly getting worse. Three reasons: (1) a growing share of very loyal old customers can hide the fact that new customers are leaving faster than ever; (2) it only counts the customers still here, ignoring everyone who already left, which makes the number look rosier than reality; (3) a flood of new signups can bury a leaky bucket of recent customers who drop off quickly.
- **Cohort groups** — Instead of one big number, group customers by the month they first signed up, then watch each group separately. That way you can see whether stores joining in March are sticking around better or worse than stores that joined in January — an answer the blended number can never give you.
- **Retention triangle (the heatmap table)** — A colour-coded table: each row is one signup-month group, each column is “how many months later.” Green cells mean customers are still around; red means they left. Reading down a column shows whether newer signups are doing better. Reading across a row shows how one group decays over time.
- **The shape of the retention curve** — Plot how many customers from a group are still active over time. A healthy business shows a steep early drop that then levels off into a flat line above zero — meaning a loyal base stays forever. If the line keeps falling toward zero with no flattening, you have a “leaky bucket” problem: no matter how many new customers you bring in, they all leave eventually.
- **Logo vs. money retention** — Logo retention counts how many store owners are still active. Revenue (dollar) retention counts whether the money you were earning is still coming in. For a low-price product like this, both numbers will move together, and it is normal and realistic to expect to keep roughly 65% of customers per year at best — not the 100%+ numbers you see quoted for expensive enterprise software.
- **Choosing the right time window** — Daily or weekly “active user” measurements make sense for apps people use every day (like social media). Print reorders happen every few months. Measuring weekly activity makes healthy, occasional-reorder customers look like they’ve left when they haven’t. The right approach is to measure activity over realistic windows (e.g. “did they order at least once in the last 90 days?”).
- **Quick Ratio (growth efficiency check)** — A simple formula: new revenue added divided by revenue lost. A result of 4 or higher means for every £1 lost, you’re gaining £4 — broadly healthy. Useful as a quick board-meeting number, but it does not replace looking at the full cohort picture.
- **Building this without expensive tools** — You do not need to buy analytics software. The same database already running the platform can produce a proper cohort retention table overnight using a short SQL script. One materialized view and a heatmap in the admin portal is enough to start.

- **Myths to ignore** — “A 5% retention improvement boosts profits 95%” is a 1990s bank statistic, not a SaaS law. “It costs 5x more to acquire than retain” has no solid source. And never multiply monthly churn by 12 to get annual churn — the real math compounds and gives a much worse (more accurate) number.

What to do with this: Stop tracking one blended churn percentage and instead build a simple cohort table (rows = signup month, columns = months since signup) directly in the existing database. Define “active” as a store that is live AND placed at least one order in the relevant window — not just “logged in.” That single change will show, for the first time, whether the product is actually retaining customers or slowly leaking them out.

Researched 2026-06-16 via a multi-agent research workflow (fan-out search + adversarial fact-check). The vendor-blog and benchmark percentages below are **hypotheses to A/B-test against our own cohort data, not guarantees** — every external SaaS number is context-dependent (ARPA, segment, buyer type) and most of the published medians come from B2B mixes that look nothing like a low-ACV, self-serve print-shop SMB. Treat the benchmarks as orientation, the *methods* (cohorts, curves, the right retention window) as the durable takeaway.

Executive Summary

Headline recommendation: Kill the single blended churn %. Make an acquisition-cohort retention triangle — built in plain SQL on our existing Postgres, cohorted by store-signup month, with “retained” defined on the print reorder cadence (not a login) — the headline retention artifact for Print-Flow-360. Do not buy ChartMogul/Amplitude/Mixpanel until that SQL view is genuinely the bottleneck.

A company-wide churn or retention number is a lagging vanity metric. It can stay flat and reassuring while the business quietly rots, because it averages over segments and cohorts that behave very differently. There are three specific ways it lies:

1. **Mix shift / Simpson’s paradox** — the blended number can *rise* even as every individual cohort retains *worse*, simply because the customer mix tilted toward stickier segments. [1][11]
2. **Survivorship bias** — measuring expansion/retention from *today’s* customers looking backward silently excludes everyone who already left. Kellblog’s illustrative example: the same data reads **111% net expansion (survivor-biased)** but **71% (unbiased, full starting cohort)**. [2]
3. **New-cohort masking** — fast-churning recent signups get hidden behind a large base of loyal old cohorts that hold the blended number steady. [1][4]

Cohorts are the diagnostic; the blended % is the symptom. For an infrequent-purchase product like print reorders, the second mistake — **using the wrong retention window** (N-day login retention) — is just as damaging: it labels perfectly healthy, low-frequency customers as “churned” and pushes the team to build pointless re-engagement nudges. [9][10]

Two non-negotiable principles for our context:

- **The retention curve must flatten above zero.** A curve that decays toward zero with no plateau is a leaky bucket / no-PMF signal — pouring our founder-led outreach and BOFU-SEO acquisition (per `ACQUISITION_CHANNELS_2026-06-15.md`) onto a leaky bucket is wasted spend. [4]
- **Do not import enterprise/high-ARPA benchmarks.** For low-ACV self-serve SMB, monthly logo churn of **2.5–5%** is *normal*/"good", and **NRR above 100% is rare** (only ~2.7% of sub-\$10/mo-ARPA products beat 100%). This is consistent with the "NRR likely below 100%" note in `CONVERSION_FUNNEL_RESEARCH_2026-06-15.md` . [3][6]

Why a single blended churn % lies

A blended retention figure averages over groups that behave very differently, so it can look healthy while the business deteriorates. Three failure modes, each with a fix:

Failure mode	What goes wrong	The fix
Mix shift / Simpson's paradox	Aggregate retention rises even as <i>every</i> segment/cohort falls, because the mix shifted toward stickier customers (and vice versa).	Never trust an aggregate. Always segment by ICP, plan, and signup cohort before believing any blended number. [1][11]
Survivorship bias	Measuring retention/expansion from <i>today's</i> customers backward excludes everyone already churned, inflating the number (111% → 71% on identical data).	Fix the starting cohort <i>in the past</i> and follow it <i>forward</i> ; keep the denominator equal to the original cohort size. [2]
New-cohort masking	A fast-churning recent cohort is hidden by a big base of loyal old cohorts that hold the blended % steady.	Read each cohort's own decay curve, not the company-wide roll-up. [1][4]

Core concepts

Acquisition cohorts

Group customers by the period they first signed up ("signed up Feb 2026"), then track each cohort forward through its own lifetime. This lets you compare January's signups at month 3 against February's signups at month 3 on equal footing, isolating the single most important growth question a blended churn number cannot answer: **are newer customers retaining better or worse than older ones?** For Print-Flow-360, cohort by **store-signup month**.

Retention triangle / heatmap

A table where each **row** is a cohort (signup month) and each **column** is months-since-signup (month 0 = signup). Cells show retention (or churn) %, color-coded green = high, red = low. It's triangular because recent cohorts haven't lived long enough to fill the right-hand columns.

How to read it:

- **Scan DOWN a column** → are successive cohorts retaining better (onboarding/product improving) or worse (quality declining)?
- **Scan ACROSS a row** → one cohort's decay curve over its lifetime.
- **Pattern direction matters:** *horizontal* red bands = a cohort-specific event (e.g. a bad acquisition campaign); *diagonal* bands = a calendar event hitting every cohort at once (outage, release, price change, theme regression); *vertical* bands = renewal/anniversary cycles.

Retention curve and the flattening “smile”

Plot retention % (y) against time-since-signup (x). Three archetypes: [4]

1. **Healthy** — steep early drop that then **flattens to a stable horizontal asymptote**. A durable base keeps coming back; the *higher* the plateau, the stronger the PMF.
2. **No-PMF “leaky bucket”** — declines continuously toward zero with no flattening. Users pass through without sticking; acquisition on top is wasted spend.
3. **Exceptional “smile”** — dips then **risers** as lapsed users return (network effects / expanding value).

The non-negotiable PMF signal is that **the curve flattens above zero**. If it doesn't, fix the product before scaling acquisition.

Logo retention vs revenue (dollar) retention

- **Logo (customer/count) retention** — what fraction of customers are still active.
- **Gross Revenue Retention (GRR)** — counts only losses (churn + downgrades), caps at 100%; the honest “are we keeping the money we had?” floor.
- **Net Revenue Retention (NRR)** — adds expansion (upsell/usage growth); can exceed 100%.

A business can lose many small logos yet hold revenue if survivors expand — so always read logo **and** dollar curves together. For low-ACV SMB with little expansion, GRR and logo retention track closely and NRR rarely beats 100%.

N-day vs unbounded (rolling) vs bracketed retention

- **N-day** — active in *exactly* period N. Strict; right for high-frequency daily products. **Wrong for us.**
- **Unbounded / rolling** — active in period N *or any period after*; counts anyone not permanently lapsed. Right for long, irregular cadence.
- **Bracketed** — custom windows (e.g. days 1–7, 8–30, 31–90), counting a return anywhere inside the bracket. Purpose-built for lumpy reorder usage.

Choosing N-day for an infrequent product makes a healthy product look dead, because most periods will correctly show zero activity. [9]

Choosing the window for an infrequent-purchase product

Reforge's rule: align the retention metric with the product's **natural usage frequency** — the cadence at which the customer re-experiences the problem you solve. [10] For print reorders that's quarters or longer,

not days. Practical approach for Print-Flow-360:

1. Measure the actual **median inter-order gap** from order data.
2. Use **bracketed or unbounded** retention over quarterly/biannual windows.
3. Define “active” as a **value event on the store’s real cadence** (an order placed/received, or a storefront still live and published), *not* a daily login.

SaaS Quick Ratio

A one-line growth-efficiency check: how much revenue you add for every dollar you lose. ≥ 4 is the oft-cited bar (Mamoon Hamid / Social Capital). [8] Useful, but it can **mask a leaky bucket** — strong new-logo inflow can produce a fine Quick Ratio even with terrible retention. Read it *alongside* cohort curves, never instead of them.

Cheap cohorts on existing Postgres

You don’t need Amplitude/Mixpanel to start. With SQL on the existing Postgres: [12][13]

1. Build a per-tenant “first activity” table (first order month, or store-go-live month) as the cohort key.
2. Generate a month spline with `generate_series`.
3. **LEFT JOIN** activity onto the spline (never INNER — an unretained customer must stay in the denominator or you silently inflate retention).
4. Compute `months_since_cohort = activity_month - cohort_month` and pivot `count(active)/cohort_size` into a triangle.
5. Materialize it as a nightly view.

Two gotchas: keep the denominator fixed to the original cohort size (survivorship guard), and filter out activity months before the cohort month.

Benchmarks

Vintage & sourcing note: The ChartMogul figures span report editions (the 2023 “Retention Report” and the 2024 “New Normal” edition) — they are labelled “ChartMogul, recent editions,” not a single 2024 dataset. The KeyBanc medians are co-published with Sapphire Ventures (2024 Private SaaS Survey: 104 companies, ~\$26M median ARR). The Lenny/Casey Winters “good/great” bands are **~2020 expert-panel** figures, not 2024 market data. **None of the high-ARPA / enterprise rows should be imported onto our print-shop SMB base** — see the low-ARPA rows for the relevant reality.

Industry-wide medians (for orientation only — NOT our targets)

Metric	Value	Confidence	Source
Median Net Revenue Retention (NRR), all SaaS, 2024	~101% (down from ~105% in 2021)	Solid	KeyBanc / Sapphire Ventures 2024 Private SaaS Survey [7]
Median Gross Revenue Retention (GRR), all SaaS, 2024	~90%, roughly stable	Solid	KeyBanc / Sapphire Ventures 2024 Private SaaS Survey [7]
Best-in-class public SaaS NRR (2025)	~120–125% (top-quartile/best-in-class)	Solid	KeyBanc 2025 SaaS Benchmark Report [7]
Share of new SaaS revenue from expansion vs new logos (2024)	~52% from expansion into existing accounts	Directional	Fullcast 2025 Revenue Benchmark Report [14]

Low-ARPA — THE relevant benchmark for Print-Flow-360

Metric	Value	Confidence	Source
NRR, low-ARPA (<\$10/mo) — top quartile	~65%; only ~ 2.7% of such businesses exceed 100% NRR	Solid	ChartMogul SaaS Retention Report [3]
Customer (logo) retention, low-ARPA (<\$25/mo) — top quartile	~65% (source: 64.7%)	Solid	ChartMogul SaaS Retention Report [3]
Monthly logo churn, B2B SMB/Mid-Market — Good	2.5%–5% per month	Directional	Lenny Rachitsky, expert-sourced [5][6]
Monthly logo churn, B2B SMB/Mid-Market — Great	<1.5% per month	Directional	Lenny Rachitsky, expert-sourced [5][6]
NRR (12-mo), Land-and-Expand SMB/Mid-Market — Good / Great	~90% Good / ~110% Great (~ 2020 expert panel)	Directional	Casey Winters × Lenny Rachitsky “What is Good Retention” [5]
User retention at 6 months , SMB/Mid-Market	~60% Good / ~80% Great (~ 2020 expert panel)	Directional	Casey Winters × Lenny Rachitsky “What is Good Retention” [5]

High-ARPA — for contrast only (do NOT adopt as targets)

The two ChartMogul high-ARPA rows use **different cohort thresholds** — the NRR row is **>\$500/mo ARPA** , the logo-retention row is **>\$1k/mo ARPA** . They are *not* the same cohort; the thresholds are stated so the rows aren’t read as one population.

Metric	Value	Confidence	Source
NRR, high-ARPA (>\$500/mo) — top quartile	~109%; ~ 41% exceed 100% NRR	Solid	ChartMogul SaaS Retention Report [3]
Customer (logo) retention, high-ARPA (>\$1k/mo) — top quartile	~86% (source: 85.8%)	Solid	ChartMogul SaaS Retention Report [3]

Growth-efficiency & bias illustrations

Metric	Value	Confidence	Source
SaaS Quick Ratio — healthy heuristic	≥4.0 (gain \$4 of MRR per \$1 lost) — rule of thumb, not a pass/fail line	Directional	Mamoon Hamid / Social Capital, via WallStreetPrep [8]
Survivorship-biased vs unbiased net expansion (illustrative)	111% (biased, survivors only) vs 71% (unbiased, full starting cohort) on identical data	Solid	Kellblog (Dave Kellogg) [2]

Formulas

Gross Revenue Retention (GRR) — the honest “money kept” floor (excludes expansion, capped at 100%):

$$\text{GRR} = (\text{Starting MRR of cohort} - \text{Churned MRR} - \text{Contraction MRR}) / \text{Starting MRR of cohort}$$

Measure off a **fixed starting cohort**, not today's survivors, to avoid survivorship bias.

Net Revenue Retention (NRR / NDR) — can exceed 100% if expansion outweighs losses:

$$\text{NRR} = (\text{Starting MRR} - \text{Churn} - \text{Contraction} + \text{Expansion}) / \text{Starting MRR}$$

For low-ACV SMB with little upsell, expect **NRR ≤ 100%** (ChartMogul: only ~2.7% of <\$10/mo products exceed 100%).

Logo (customer) retention rate:

$$\text{Logo retention (period N)} = (\text{Customers from cohort still active at period N}) / (\text{Total customers in ORIGINAL cohort})$$

The denominator is the **original cohort size and stays fixed across all N** — this is the survivorship guard. Use **LEFT JOIN** in SQL so lapsed customers remain in the count.

SaaS Quick Ratio:

$$\text{Quick Ratio} = (\text{New MRR} + \text{Expansion MRR}) / (\text{Churned MRR} + \text{Contraction MRR})$$

≥ 4 = healthy growth efficiency. Caveat: a high ratio driven by new logos can hide weak retention — pair with cohort curves.

Unbounded (rolling) retention — the right metric for low-frequency products:

Unbounded retention (period N) = (Users active in period N OR any later period) / (Cohort size)

Bracketed retention — purpose-built for irregular reorder cadence:

Bracket retention = (Users with ≥ 1 value event inside window [a,b]) / (Cohort size)
e.g. windows [1-30d], [31-90d], [91-180d], [181-365d] around the observed median inter-order gap

Churn ↔ **retention identity** (true only within a single cohort/period):

Period retention = 1 - Period churn

Do **NOT** annualize monthly churn by multiplying $\times 12$. Churn compounds:

Annual retention = $(1 - \text{monthly churn})^{12}$

What this means for Print-Flow-360

Concrete, prioritized, and scoped to a small non-technical SMB customer base with limited eng time.

Instrument (in priority order)

1. **Ship ONE materialized Postgres view + a simple heatmap in the admin portal first.** Do NOT buy ChartMogul/Amplitude/Mixpanel until the cohort base is large enough that the SQL view is genuinely the bottleneck.
2. **Make the acquisition-cohort retention triangle the headline retention artifact.** Rows = store-signup month, columns = months since signup. Cohort by **month** (not week) — the install base is small and weekly cohorts will be too noisy.
3. **Define “active/retained” on the print product’s real cadence, not a login.** A store is retained for a period if it (a) stays published/live **AND** (b) places or receives ≥ 1 order within the window.
4. **Use UNBOUNDED or BRACKETED retention, never N-day.** First measure the actual median inter-order gap from order data, then set brackets around it (e.g. 0–30d, 31–90d, 91–180d, 181–365d).
5. **Track logo AND dollar retention separately.** Expect GRR and logo retention to track closely and NRR to sit at/below 100%.
6. **Instrument a monthly Quick Ratio** = $(\text{new} + \text{expansion MRR}) / (\text{churned} + \text{contraction MRR})$ as a one-line board metric, target ≥ 4 — but never let it substitute for the cohort curve.

SQL hygiene to bake into the cohort view:

- **LEFT JOIN** activity to a **generate_series** month spline (**never INNER** — keeps lapsed stores in the denominator).
- Keep each cohort's denominator **fixed to its original size** (survivorship guard).
- Filter out activity months **earlier than** the cohort month.
- Materialize **nightly, per tenant**.

Read it every month, two ways

- **DOWN columns** → are newer cohorts retaining better than older ones? (Is onboarding improving?)
- **ACROSS rows** → each cohort's decay curve.
- **Diagonal red bands** = a platform-wide event (outage, pricing change, theme regression). **Horizontal red bands** = a bad batch of signups from one campaign.

Tie it to the existing North Star

The conversion-funnel doc set the North Star as **store live + first order in 7 days**. Cohort the **activation event** and check whether stores that hit activation in week 1 show the flattening curve, while non-activated stores leak to zero. This validates (or kills) the activation hypothesis with real cohort evidence.

What to A/B test (hypotheses, not guarantees)

- **Activation lever:** does pushing stores to *first order in 7 days* (the funnel doc's bet) measurably raise the cohort plateau vs control? Cohort both arms and compare flattened retention.
- **Onboarding changes:** ship onboarding improvements one cohort at a time, then scan DOWN the column to see if successive post-change cohorts retain better.
- **Re-engagement nudges:** only worth building if bracketed retention (not N-day) shows real lapse inside a *natural-cadence* window. Test a nudge against a hold-out and measure return rate inside the next bracket.

What to ignore

- **A single blended churn/retention %** as a health signal — it lies via mix shift and survivorship.
- **NRR > 100% as a target.** For our low-ACV SMB base it's unrealistic (ChartMogul: <\$10/mo top-quartile NRR ~65%). The realistic growth lever is **more new logos + per-store order-volume/product expansion**, not seat upsell.
- **Enterprise/high-ARPA benchmarks** (the >\$500/mo and >\$1k/mo rows, "negative churn," "best-in-class 120% NRR"). They describe a different business.
- **Daily/weekly login retention** as a signal for an infrequent-reorder business.

Folklore & mis-attributed stats — flagged explicitly

These recur in SaaS-metrics writing and are wrong, mis-sourced, or mis-applied. **Do not present any of them as a hard SaaS law.**

- **“A 5% increase in retention boosts profits 25–95%.”** From Bain / Fred Reichheld (Reichheld & Sasser, “Zero Defections”), **not** an original HBR study, and the 25–95% is a wide **1990s cross-industry range**, not a SaaS-specific law (the 85% figure came from one bank’s branch system). Cite carefully; never present “95%” as the expected number.
- **“Acquiring a new customer costs 5x (or 7x, or 25x) more than retaining one.”** Folklore with no robust primary source; the multiplier varies wildly and is often invented. Directional reminder only, not a benchmark.
- **A flat blended NRR/churn number presented as proof of health** is itself a folklore trap. Simpson’s paradox means the aggregate can improve while every cohort worsens (and vice versa). Demand the segmented/cohort view before believing any blended figure. [11]
- **Survivorship-biased retention reported as if unbiased.** Kellblog’s identical-data example: 111% (biased) collapses to 71% (unbiased). Be skeptical of any NRR computed from “current customers looking back.” [2]
- **“Quick Ratio ≥ 4 = good company.”** A useful heuristic (Mamoon Hamid / Social Capital), **not a validated threshold**; a high ratio can mask a leaky bucket fueled by new logos. Don’t treat 4.0 as a pass/fail line. [8]
- **“Good SaaS churn is 5–7% annually” / “best-in-class is negative churn.”** Lifted from enterprise/high-ARPA contexts; do **not** apply to low-ACV self-serve SMB, where monthly logo churn of **2.5–5%** is normal/“good” and NRR > 100% is rare.
- **Annualizing monthly churn by multiplying $\times 12$.** A common arithmetic error — churn **compounds**; use $(1 - \text{monthly churn})^{12}$.
- **“<40% Day-1 retention guarantees <15% 12-month retention”** (and similar precise D1 → annual coupling claims) is fabricated/unsourced. The defensible version is only that *early value delivery correlates with stronger long-term retention*.

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13. Stripe — SaaS cohort analysis as a framework for improving churn, expansion, and revenue. <https://stripe.com/resources/more/saas-cohort-analysis>
14. Fullcast 2025 Revenue Benchmark Report (52% of new revenue from expansion, 2024 data).

Related internal docs: [readme/CONVERSION_FUNNEL_RESEARCH_2026-06-15.md](#) (North Star = store live + first order in 7d; reverse trial), [readme/ACQUISITION_CHANNELS_2026-06-15.md](#) (founder-led outreach + print communities + BOFU SEO). This doc is research/strategy only — no code changes.

Funnel Analytics & Event Taxonomy Design

Source: readme/research/metrics-analytics/METRICS_FUNNEL_EVENT_TAXONOMY_2026-06-16.md

IN SHORT — KEY TAKEAWAYS

This document is a practical guide for tracking whether the Print-Flow-360 platform is actually helping store owners succeed — specifically, whether a new store places its first real order within 7 days of signing up. It tells you exactly which events to track, how to name them, and which tool to use, so the team gets reliable numbers instead of a confusing pile of data nobody trusts.

The main parts explained simply:

- **What a funnel is** — A funnel counts how many people complete each step of a process in order (for example: sign up → publish store → receive first order). Each step shows you where people are dropping out so you can fix the right thing.
- **Conversion window** — The time limit you set for someone to complete all funnel steps. This document recommends 7–14 days for the “first order” goal — long enough for a real store owner who signs up on a Friday and orders the following week, short enough to keep the numbers honest.
- **Event naming rules (taxonomy)** — A set of simple rules for naming every action you track, so your data doesn’t get messy over time. The rule is: `object_action` in lowercase with underscores, past tense (for example, `order_received`, `store_published`). Variable data like “what type of product” goes as a detail on the event, not baked into the name itself.
- **Identity stitching** — Making sure one real person is counted as one person in your analytics, even before they log in. If you skip this step, your numbers are silently wrong — no error message tells you, the data just lies quietly.
- **Server-side vs browser-side tracking** — Tracking from the server (your backend) is more reliable than tracking in the browser, because browser-based tracking can be blocked by ad-blockers. The most important events — sign-ups, orders, payments — should always be tracked from the server.
- **Tool recommendation (PostHog)** — PostHog is the recommended analytics tool. It has a generous free plan, tracks clicks and page views automatically without extra coding, and includes session recordings and feature flags all in one place.
- **The 14-event tracking plan** — A short, concrete list of the only events worth tracking right now. Keeping the list small keeps the data trustworthy. Adding events should require a team review.
- **Governance (keeping it tidy)** — Rules to stop the tracking plan from becoming a mess: every event needs an owner, changes go through a checklist before shipping, and nothing is renamed or deleted without a clear migration plan.

What to do with this: Use PostHog and start by tracking just these three events from the server:

`signup_completed`, `store_published`, `first_order_received`. Review weekly how many stores reach all three within 14 days. That single number tells you more than any dashboard full of clicks.

Researched 2026-06-16 via a multi-agent research workflow (parallel fan-out + adversarial verification pass). This is an internal design doc, not a vendor pitch. **Vendor-blog percentages are hypotheses to A/B-test against our own data, not guarantees** — every benchmark below carries a Confidence rating, and the most-quoted SaaS-metrics figures (ad-blocker data-loss, free-tier allowances) are explicitly flagged as drifting or unaudited. Free-tier and per-event pricing numbers are 2026 snapshots and **must be re-verified before you commit to a tool** — one of them (Mixpanel) already drifted during this very check.

Executive Summary

Headline recommendation: Adopt PostHog, fire all money/lifecycle events server-side from the Laravel service layer, identify on the tenant `uuid`, and ship a deliberately small ~14-event tracking plan whose single job is to measure ONE funnel — `signup_completed` → `store_published` → `first_order_received` within 7–14 days. Do not build a Postgres-only analytics product, and do not hand-instrument clicks.

We are a low-ACV, self-serve SMB print SaaS selling to non-technical print-shop owners. The failure mode for a team like ours is not “too little data” — it is **event sprawl, fragmented identity, and a beautiful dashboard nobody can trust**, all built before the one funnel that matters is even measured. The single North Star from `CONVERSION_FUNNEL_RESEARCH_2026-06-15.md` — *stores with an order within 7 days* — maps to exactly one event (`first_order_received`) on exactly one identity grain (the tenant). Everything in this doc exists to measure that reliably and cheaply.

Three opinionated calls:

1. **Server-side for truth, autocapture for everything else.** Our highest-value events (signups, orders, payments) already run in the Laravel service layer. Firing them there is simultaneously *more reliable* (un-blockable) and *less work* than wiring browser instrumentation. Client-side autocapture covers low-value clicks/pageviews for free.
 2. **One identity grain: the tenant.** The North Star counts *stores*, so the tenant `uuid` is the primary analytics identity, with `user_id` carried as a property. Two-moment identity (init anonymous on load, `identify()` on auth) is non-negotiable — the failure here is silent.
 3. **Measure before you optimize.** Do not touch the storefront micro-funnel until the activation funnel is instrumented, segmented, and reviewed weekly. Premature funnel optimization hides the bigger problem.
-

1. Funnel concepts (the parts people get wrong)

What a funnel actually counts

A funnel is an **ordered set of events measured as the % of unique users who reach each step within a conversion window**. Two things people routinely conflate:

- Funnels count **unique users, not raw events**. A user converts once per attempt. The “Insights / Totals” view uses a *different denominator* (raw event counts) and will mislead you if you read it as a funnel rate [6].
- Per-step rate **localizes the leak**; the overall rate is the product of the step rates.

```
step_conversion = unique_users_reaching_step / unique_users_reaching_prior_step
overall         = unique_users_completing_final_step / unique_users_entering_step_1
```

Conversion window — a modeling choice, not a detail

The conversion window is the max time allowed between the first and last funnel step. Set it too short and you under-count slow-but-real conversions; too long and you over-credit unrelated later sessions to the funnel. For **Print-Flow-360 activation, a 7–14 day window matches the “first order in 7 days” North Star** — long enough to capture a print-shop owner who signs up Friday and places a real order the following week, short enough to stay honest.

Folklore correction (this was the most-wrong claim we checked). A widely repeated phrasing — “*Mixpanel’s conversion window runs from a 2-second floor to 90 days*” — is **fabricated by conflation** and should never be cited. Two separate facts get mashed together:

- The **2-second figure is the near-simultaneous-step grace period** (see below), *not* a window floor.
- The **maximum conversion window is 366 days** (12 months / 52 weeks), or 12 sessions for session-based windows, with a **default of 7 days** — *not* 90 days [1a]. If you have seen “2s to 90d” in a slide deck, it is wrong on both ends.

Strict vs loose (flexible) ordering

Ordering mode	Means	Use for
Strict	Steps occur in exact sequence; out-of-order actions break the funnel	Linear flows: checkout → payment, the go-live wizard
Loose (flexible)	Steps occur in order, but other actions are allowed in between	Exploratory journeys: browse → configure → publish

Choosing wrong **counts the wrong population entirely**. Our storefront purchase funnel is linear → strict. Our activation funnel allows wandering (a user explores products, tweaks themes, leaves, comes back) →

strict-ish/loose. Mixpanel additionally applies a **2-second grace period**: consecutive steps with timestamps within 2 seconds of each other are interchangeable, so near-simultaneous events (e.g. a page-view firing microseconds before a click) don't falsely fail the order check [1].

Multi-path funnels & segmentation — the two silent leak-fakers

- **Multi-path**: products have more than one route to value. Forcing one linear funnel through a step many users *legitimately skip* inflates that step's drop-off and produces a **false leak**. Fix: make questionable steps optional, or model separate funnels per path.
- **Segmentation**: a blended funnel hides the leak. **Always break the same funnel down** by acquisition channel, persona/vertical, device, and signup cohort to localize *where* and *for whom* it leaks [2][3]. For us: segment by `store_vertical` and `first_touch_source` at minimum.

2. Funnel benchmarks & platform facts

Metric	Value	Confidence	Source
Mixpanel conversion-window max	366 days (12 months / 52 weeks) or 12 sessions; default 7 days	Solid (corrected from a fabricated “90 days”)	Mixpanel Docs — Funnels Advanced [1a]
Mixpanel near-simultaneous-step tolerance	2-second grace (steps within 2s are interchangeable)	Verified	Mixpanel Docs — Funnels FAQ [1]
Mixpanel identity-merge lag	up to 24 hours (a ceiling, not a typical value) between anon and known cohorts	Verified	Mixpanel Docs — Funnels FAQ [1]
Amplitude tracking-plan guideline	~10–200 events; ≤20 properties/event — <i>recommendation, not a hard limit</i>	Solid	Amplitude via Digital Applied [7]
GA4 hard limits	event names ≤40 chars (must start with a letter), 25 params/event, 50 event-scoped custom dimensions	Verified	GA4 platform limits via Digital Applied [7]

The “2-second floor to 90 days” range is gone on purpose. We split it into its two real facts above. The 2-second number lives *only* in the grace-period row; the window max is 366 days, default 7. Cite [1a] (Funnels Advanced) for the window — the Funnels FAQ does **not** document the range.

3. Event taxonomy — naming that won't rot

Object-Action (`verb_noun` → `noun_verb`) naming

An event is **object (noun) + action (verb)** — e.g. `order_received` , `product_added` . The Amplitude/Mixpanel/PostHog house style is **lowercase, underscores, past tense**.

Folklore flag. The “Object-Action framework” has **no single canonical inventor** — it's a shared convention documented across Amplitude [6], Mixpanel, and PostHog [8]. Don't attribute it to one vendor as proprietary IP.

The four naming axes — lock once, enforce forever

Axis	Our choice
Casing	<code>snake_case</code> (casing drift <i>alone</i> fragments metrics into duplicates)
Format	<code>object_action</code> so related events cluster alphabetically (<code>order_received</code> , <code>order_refunded</code>)
Tense	past simple (<code>added</code> , not <code>add</code> / <code>adding</code>)
Vocabulary	approved verb list (below)

PostHog's default restricted verb list (a strong default, not hard enforcement): `view`, `click`, `submit`, `create`, `add`, `update`, `delete`, `remove`, `start`, `end`, `cancel`, `fail`, `send`, `invite`, `generate` [8]. (*Confidence: Solid.*)

Two rules that prevent taxonomy explosion

1. **Event names are fixed strings — never build a name from a variable.** `order_${status}` creates a new “event” for every status and detonates the taxonomy. Variable data goes in **property values**, not names.
2. **Events vs event-properties vs user-properties:**
 - **Event** = a discrete action (`product_added`).
 - **Event property** = context of *that occurrence* (`product_type` , `price_set`).
 - **User property** = persistent attribute of the person/store (`plan` , `store_vertical` , `is_b2b`).
 - If you're tempted to split `product_added_business_cards` vs `product_added_flyers` , those are **two property values of one event**. Properties prevent sprawl.

Event sprawl avoidance

Track **outcomes tied to business goals, not every interaction**. Over-tracking inflates volume and obscures patterns. Let **autocapture** handle low-value clicks/pageviews for free; hand-instrument **only lifecycle events**. For an SMB team, **stay near 12–15 events**.

Folklore flag. Amplitude's "10–200 events / ≤ 20 properties per event" is **functional guidance, not an enforced platform limit** [7] — don't cite it as a rule that will reject your data. The **inverse** is true for GA4: its 40-char name and 25-param limits are **real, load-bearing constraints** — exceed them and data is dropped *silently*. If we ever dual-ship to GA4, names must fit.

Governance — the part that actually fails

Taxonomies rot from **no enforcement gate, not from a bad spreadsheet**. The highest-cost decay modes throw **no error**: silent identity fragmentation, and orphaned events whose author left the team. Minimum viable governance:

- A **living tracking plan** checked into the repo with a **named owner per event**.
- A **PR checklist** for any tracking change.
- **Version labels** (`proposed` / `active` / `deprecated` / `removed`) — **never silent renames or deletes**; deprecations get a migration plan.

4. Identity — the error that throws no error

Identity stitching (anonymous → identified)

Every browser gets an **anonymous/device id**. On signup/login you call `identify(user_id)`, which **binds the anon id to the real id and retroactively merges prior anonymous events** into that profile [9][10].

The two-moment identity rule (non-negotiable)

1. **Init identity on first load** — bind the anonymous session.
2. **Call `identify()` after login/signup** — merge anon → known.

Skip either and you get the **highest-cost, error-free failure**: cohorts are silently wrong because one human is split across multiple ids, or many humans collapse onto one shared device. There is no exception, no log line, no red dashboard — just quietly false numbers. (Note Mixpanel's *up to 24-hour* merge lag [1] when reconciling anon vs known cohorts — don't panic at a same-day discrepancy.)

`$set` vs `$set_once`

- `$set` **overwrites** — use for **mutable** attributes (`plan`, current subscription state).
- `$set_once` **preserves** — use for **immutable acquisition context** (`first_touch_source`).
- **Overwriting acquisition source destroys attribution permanently**. First-touch is `$set_once`, always.

Multi-tenant identity grain (Print-Flow-360 specific)

We have **two grains: tenant/store and individual user**. Because the North Star is *stores with an order in 7 days*:

- **Primary analytics identity = tenant `uuid` .**
- Carry `user_id` as a **property**, not the identity.
- **Never expose raw uuids in user-facing labels** (per [CLAUDE.md](#) §0 — derive a human store name).

5. Server-side vs client-side capture

Client-side events pass through the browser, where **ad-blockers, ITP, and JS failures drop them**. Server-side events go straight from the backend and **cannot be blocked**. The cruel part: **the most-lost events are the highest-value ones** — signups, orders, payments [11][12][13].

Our model is hybrid: client-side **autocapture** for low-stakes data (pageviews, clicks, inputs — free, no manual work); **server-side** for money/lifecycle truth. Our lifecycle/revenue events already execute in the Laravel service layer, so server-side capture is both more reliable *and* less work.

```
loss_pct = (server_event_count - client_event_count) / server_event_count
```

Treat the **server count as source of truth**; the gap approximates blocker/ITP/JS loss. **Directional, not exact.**

Metric	Value	Confidence	Source
Ad-blocker adoption	~29.5% global, ~33% US (Q2 2025)	Solid	Backlinko (GWI data) [14]
Ad-blocker analytics event-loss	~ 25–40% of client-side traffic <i>in some industries</i>	Directional — vendor-stated, not independently audited	Digital Applied / Cometly / Snowplow [7][11][13]

Folklore flags (server-side tracking justification — use directionally only):

- “Ad-blockers cause 25–40% analytics data loss” — ad-blocker **adoption** (~29–33%) is well-measured [14], but the **event-loss** figure is vendor-stated and explicitly **unaudited**. Use it to justify server-side capture (directionally true), **not as a precise target**. (The range is 25–40% per the cited source — not the “20–40%” some decks quote.)
- “You lose a \$5,000 sale every time a pixel is blocked” — vendor illustration, **not data**.
- “One fintech lost exactly 40% of conversions to ad-blockers” — single uncited anecdote; directional at best.

6. Tool selection

Recommendation: PostHog. Generous free tier covers SMB volume, **autocapture** handles low-value clicks/pageviews for free, it **bundles funnels + session replay + feature flags**, and it’s **self-hostable** (a

real plus for EU / multi-tenant data posture). The feature flags also operationalize the A/B tests in §7 with no extra vendor.

Do NOT build a Postgres-only analytics product. We have limited eng time; reinventing funnels, identity merge, and retention math is exactly the sprawl trap this doc warns against. `mcp__postgres-test` exists for *testing the app*, not as an analytics warehouse.

Tool	Free-tier (2026 snapshot — re-verify before committing)	Per-event beyond free	Confidence	Source
PostHog	~1M analytics events/mo + 5k session replays + 1M feature-flag requests, unlimited seats	~ \$0.00005 /event (first paid 1–2M tier; cheaper with volume), ~\$0.005/session replay	Solid (free tier) / Directional (price)	AgentDeals 2026 / PostHog [16][15]
Amplitude Starter	50k MTUs / 10M events/mo (MTU is the binding gate), ~1,000 replays, 12-mo retention	—	Solid	AgentDeals 2026 [16]
Mixpanel	~ 1M events/mo (unlimited seats)	—	Directional	AgentDeals 2026 [16]

Folklore / drift flags (this is exactly where free-tier folklore rots):

- **PostHog per-event price was overstated ~6×** in the folklore we inherited (a quoted “\$0.00031/event”). The real first-paid-tier analytics price is **~\$0.00005/event** [16]; session replay at ~\$0.005/recording is confirmed.
- **Mixpanel switched to event-based pricing in Feb 2026.** The old “**1M MTU**” free-tier phrasing is now **wrong for new accounts** — it’s **~1M events/month**; legacy MTU plans are grandfathered. This is the exact drift our caveat warned about, and it happened.
- **Amplitude’s binding constraint is the 50k MTU gate**, not the headline “10M events” — most teams hit MTUs first.
- **All three free tiers drift frequently and sources already disagree.** Re-verify before you commit budget.

7. What this means for Print-Flow-360

This section is the operational contract. It directly **operationalizes** `CONVERSION_FUNNEL_RESEARCH_2026-06-15.md`: the North Star “stores with an order in 7 days” is the `first_order_received` event; the behavioral upgrade trigger is `card_added` with `trigger: post_first_order`. It respects `ACQUISITION_CHANNELS_2026-06-15.md` by making `first_touch_source` (`$set_once`) the primary segmentation axis.

Instrument (do this)

1. **Fire all server-side lifecycle/revenue events from the Laravel service layer** — they already run there, so this is both more reliable than blockable client events *and* less work: `signup_completed`, `product_added`, `price_set`, `store_published`, `order_received` / `first_order_received` / `reorder_received`, `card_added`, `subscription_changed`.
2. **Adopt PostHog.** Autocapture for low-value events; self-hostable; bundles funnels + replay + flags.
3. **Wire identity in two moments:** init anon id on first load, call `identify(tenant_uuid)` at signup/login, `$set_once` the `first_touch_source`. Tenant `uuid` = primary grain, `user_id` = property, **never raw uuids in labels**.
4. **Build the activation North Star funnel:** `signup_completed` → `store_published` → `first_order_received`, **strict-ish ordering, 7–14 day window**, segmented by `store_vertical` and `first_touch_source`. **Review the worst-converting step weekly.**
5. **Build a second strict, short-window storefront purchase funnel:** `landing_viewed` → `product_added` → `checkout` → `order_received` (checkout is linear → strict).
6. **Encode variation in properties, not events** — `product_type`, `customer_type` (`guest` / `b2b`), `source` (`demo_seed` / `manual`). This connects to the existing **B2B** and **demo-seed onboarding** work without minting new events.

The starter tracking plan (~14 events — checked in, owned, PR-gated)

Snake_case, past-tense, object_action. **S** = server-side (truth), **C** = client-side (autocapture/context). Every event has a named owner; add a new event **only when an existing one cannot answer a real question**.

Event	Side	Key properties	User properties touched	Funnel	Owner
landing_view ed	C	path , first_touch_source	\$set_once first_touch_source	purchase	Growth
signup_compl eted	S	source , store_vertical	\$set plan , \$set store_vertical	activation	Growth
store_config ured	S	step , fields_completed	—	activation	Onboarding
product_adde d	S	product_type , source (demo_seed / manual), price_set (bool)	—	activation	Catalog
price_set	S	product_type , strategy	—	activation	Pricing
store_publis hed	S	time_since_signup_hours , product_count	—	activation	Onboarding
checkout	S	cart_value , customer_type (guest / b2b)	—	purchase	Commerce
order_receiv ed	S	product_type , customer_type , amount , currency	—	purchase	Commerce
first_order_ received	S	time_since_publish_hours , store_vertical	\$set activated true	North Star	Commerce
reorder_rece ived	S	days_since_last_order	—	retention	Commerce
card_added	S	trigger (post_first_order / manual)	—	monetization	Billing
subscription _changed	S	from_plan , to_plan , reason	\$set plan	monetization	Billing
support_requ ested	S	channel , topic	—	support	Support
design_save d	C/S	template_used (bool)	—	retention	Studio

Time-to-value (surface as properties, not new events)

```
ttv = timestamp(first_order_received) - timestamp(signup_completed)
```

Carry `time_since_signup_hours` / `time_since_publish_hours` as event properties. This is our core activation latency and the reason the window is 7–14 days.

```
activation = stores_reaching_store_published AND first_order_received within 7-14d / signups
```

The **one funnel to review weekly**, segmented by `store_vertical` and `first_touch_source`.

Ignore (do NOT do this — it's the sprawl trap)

- **Do NOT hand-instrument** per-button clicks, scroll depth, heatmaps, or per-field form analytics. **Autocapture + session replay cover these for free**; manual instrumentation here is the sprawl.
- **Do NOT optimize the storefront micro-funnel** before the activation funnel is measured and healthy — premature funnel optimization hides bigger problems [8].
- **Do NOT build a Postgres-only analytics product.**
- **Do NOT build event names from variables** (`order_${status}`) or split one action into per-value events.

A/B test (on our own funnel only — these are hypotheses, not facts)

1. **Activation conversion-window length** — does 7 vs 14 days materially change the measured activation rate, and which better predicts retained stores?
2. **Behavioral vs calendar** `card_added` **trigger** — `trigger: post_first_order` (behavioral) vs a calendar-day prompt.
3. **Demo-seed vs blank-start** — effect of a seeded demo store on the `product_added → store_published` step (use the `source: demo_seed` property, no new event).

Governance gate (the PR checklist)

Any tracking change must, in the PR:

- [] Use a **fixed-string** event name (no variables); `snake_case`, past-tense, approved verb.
- [] Type every property; encode variation as **property values**, not new events.
- [] Set the **owner** field.
- [] For changes: apply a **version label** (`proposed` / `active` / `deprecated` / `removed`) — **no silent renames**; deprecations include a migration plan.
- [] Confirm identity: anon-init present, `identify(tenant_uuid)` on auth, `first_touch_source` via `$set_once`.

Sources

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<https://amplitude.com/blog/three-steps-to-increase-your-conversion-rate-using-funnels>

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15. Medium (Justine) — Amplitude vs Mixpanel vs PostHog: The Honest 2026 Comparison: <https://talking-tech-with-j.medium.com/amplitude-vs-mixpanel-vs-posthog-the-honest-2026-comparison-25696721d9c5>
16. AgentDeals — Analytics & Product Analytics Free Tier Comparison 2026 (re-verify; drifts): <https://agentdeals.dev/analytics-free-tier-comparison-2026>

Maintenance: free-tier and per-event pricing in §6 are 2026 snapshots and drift often — re-verify before budgeting. If you change the tracking plan in §7, update this doc in the same PR (the plan is the source of truth, not a screenshot of the PostHog UI).

MRR/ARR Mechanics: Expansion, Contraction, Net Revenue Retention

Source: [readme/research/metrics-analytics/METRICS_MRR_ARR_NRR_2026-06-16.md](#)

IN SHORT — KEY TAKEAWAYS

This document explains how to measure the health of a subscription software business in honest, accurate numbers. It matters because using the wrong definition — even once — causes every other number (growth, retention, valuation) to lie, and decisions based on wrong numbers lead the business in the wrong direction.

The main parts explained simply:

- **MRR and ARR (what “recurring revenue” actually means)** — MRR is the monthly subscription fee income, nothing else. An annual plan is spread evenly: a \$600/year plan counts as \$50/month, not \$600 in one month. ARR is just MRR multiplied by 12. Print order revenue, setup fees, and one-time charges must never be added in — doing so makes the business look bigger than it is and breaks every other calculation.
- **The five movements of MRR (where growth actually comes from)** — Each month, MRR changes for five reasons: new customers joining, existing customers upgrading, old customers coming back, existing customers downgrading, and customers cancelling. Tracking all five separately shows exactly which problem to fix. Most teams only watch “new” and “cancelled” and miss the warning signs in between.
- **Contraction as an early warning sign** — When a customer drops to a cheaper plan, that often means they are about to cancel entirely. Watching this number weekly gives time to reach out and save the relationship before the full cancellation happens.
- **NRR — how well you keep the money you already have** — Net Revenue Retention measures how much subscription income stays from existing customers, including any upgrades. Above 100% means existing customers are spending more over time. Below 100% means you must keep signing new customers just to stay flat. For small-business software tools like this one, around 97% is the normal, healthy level — not 120% (that is an enterprise-software target and does not apply here).
- **GRR — the honest leak test** — Gross Revenue Retention measures the same thing but ignores upgrades, showing only the damage from cancellations and downgrades. It can never be above 100%. If NRR looks fine but GRR is falling, it means a few big upgrades are hiding a lot of cancellations — a sign of real trouble underneath.
- **GMV — a health signal, not revenue** — GMV is the total value of print orders that flow through the platform. It is not the platform’s revenue. But it is the single best sign that a store is active and unlikely to cancel. Track it separately as a health indicator, never mix it into subscription revenue numbers.
- **Annual plan accounting** — When a customer pays a full year upfront, that cash cannot all count as income in one month. It must be spread across 12 months in the books. The upfront cash is still a good thing (better for cash flow and customers cancel less), but the revenue recognition must stay spread out.
- **Common myths to ignore** — “Every SaaS should hit 120% NRR” is an enterprise benchmark that does not apply to small-business tools. “Quick Ratio of 4 is a must” is a rule of thumb that means nothing when the customer base is still small. Snowflake’s 158%+ NRR figures are from a massive enterprise data-warehouse company — irrelevant as a target here.

What to do with this:

Track the five MRR movements every week (new, upgraded, reactivated, downgraded, cancelled) and keep subscription income completely separate from order/transaction income. Watch for stores that downgrade — they are likely about to cancel — and reach out early. Accept that ~97% retention is a healthy target for this type of business; do not chase enterprise benchmarks.

Researched 2026-06-16 via a multi-agent workflow (research + adversarial fact-check). This doc is an internal reference for how Print-Flow-360 should define and instrument its recurring-revenue metrics.

Vendor-blog and survey percentages here are hypotheses and segment medians to benchmark against and A/B-test toward — not guarantees for our business. Every number carries a Confidence label; treat anything below “Solid” as directional. The most damaging mistakes in this domain are definitional (blending order revenue into MRR, chasing the wrong NRR target), not arithmetic — this doc leads with those.

Executive Summary

Headline recommendation: keep ONE clean recurring-revenue base — MRR/ARR = subscription plan fees only (Starter/Growth/Pro, normalized monthly) — report platform GMV and any take-rate revenue on a separate line, and accept that ~97% NRR is healthy for our low-ACV SMB segment. Do not chase 120% enterprise NRR; it's the wrong target and will mislead strategy.

Print-Flow-360 earns two fundamentally different kinds of money: predictable **subscription revenue** (the plan fees — this is the MRR/ARR base that earns high valuation multiples) and **transactional order revenue** tied to the dollar value of print jobs flowing through stores (GMV). These obey different math and must never be summed into a single “ARR” number. Blending them is the single most common and most damaging recurring-revenue misstatement, and investors discount or flag it.

For a self-serve, low-ACV SMB tool sold to non-technical print-shop owners, the realistic growth model is **lots of small new logos + meaningful churn + modest expansion** — not the expansion-led “negative churn” story enterprise SaaS tells. That means:

- The **MRR movement waterfall** (New / Expansion / Reactivation / Contraction / Churn) is the core weekly artifact — it tells you *which* lever is moving, which is almost certainly retention/activation (consistent with the funnel research’s North Star of “store live + first order in 7 days”).
- The **one durable expansion lever** for this audience is usage-based tier graduation: as a store’s order volume grows, nudge Starter → Growth → Pro. This is how platform success (GMV) becomes subscription expansion MRR.
- **GMV is your best health/leading-churn indicator** — a store doing order volume rarely churns — but it is *not* recurring revenue.
- With limited eng time and a sub-scale paying base, **track a short weekly list and skip the precision metrics** (Quick Ratio, cohort NRR/GRR) until you have a few hundred paying stores.

1. The core definitions (get these right or everything downstream is wrong)

MRR and ARR

MRR is normalized, amortized monthly subscription revenue — only money that contractually keeps coming on a subscription cadence. Annual and quarterly plans are normalized to a monthly figure; a \$600/yr plan contributes **\$50/mo** MRR recognized evenly across the term, *not* \$600 in the signup month. [1][13]

```
MRR = Σ (normalized monthly subscription value of each active subscription)
    annual plan → annual price / 12
    quarterly   → quarterly price / 3
ARR = MRR × 12
```

ARR is a run-rate snapshot of current MRR annualized — NOT the sum of annual bookings or annual cash collected. [13]

For Print-Flow-360, **MRR/ARR = ONLY the plan subscription fees** (Starter/Growth/Pro). Per-order print/transaction revenue, setup fees, one-time design work, and usage overages are *never* in the base — even though order revenue is the bulk of cash flowing through the platform.

What does NOT count as MRR

Excluded entirely, because they lack the predictability ARR requires [1][13]:

Excluded item	Why
One-time fees (setup, onboarding, implementation)	Not recurring
Usage-based / overage / consumption charges	Variable, unpredictable
Professional services / ad-hoc custom design	Billed ad-hoc, not on a cadence
Hardware / physical-goods sales	Not subscription
Time-limited promo discounts	Shouldn't inflate or deflate the base artificially
Per-order print / transaction revenue	Non-recurring — the #1 thing to keep out of our MRR

Including any of these inflates MRR and corrupts every retention and Quick-Ratio calculation downstream.

2. The MRR Movement Waterfall

The waterfall decomposes period MRR change into five movements — the single most diagnostic recurring-revenue view, because it tells you *which lever* is moving, not just net growth. [2][3]

```
End MRR = Start MRR
          + New MRR      (brand-new customers)
          + Expansion MRR (existing customers upgrading tier / adding seats / add-ons)
          + Reactivation MRR (previously-churned customers returning to paid)
          - Contraction MRR (existing customers downgrading but staying)
          - Churned MRR    (customer cancels their last/only subscription – full loss)
```

```
Net New MRR = New + Expansion + Reactivation - Contraction - Churned
```

The five movements must **reconcile exactly** from start-MRR to end-MRR. Convention (e.g. ChartMogul) renders New/Expansion/Reactivation above the \$0 line and Contraction/Churn below. [2][4]

Underused middle of the waterfall. Most teams obsess over New and Churn and ignore Reactivation and Contraction. [4]

- **Reactivation** (winning back lapsed stores) is often cheaper than new acquisition — fits the founder-led-outreach motion from the acquisition research.
- **Contraction is an early-warning signal that precedes churn** — a store downgrading is frequently a store about to leave. Watch it weekly to get a head start on a save-motion before the full cancel.

3. Retention metrics: NRR and GRR

Net Revenue Retention (NRR / NDR)

How much recurring revenue you keep from **existing customers** over a period, **including expansion** but **excluding new logos**. [5][6]

```
NRR = (Starting MRR + Expansion - Contraction - Churn) / Starting MRR
      measured on the SAME cohort, EXCLUDING new logos
      (usually a trailing-12-month cohort to smooth noise)
```

- **>100%** = the existing base grows on its own — the “leaky bucket that refills itself,” the holy grail of efficient SaaS.
- **<100%** = you must win new logos just to stand still.

SMB / low-ACV SaaS structurally sits below 100% because the upsell ceiling is limited. This is a *segment reality, not a failure* — Print-Flow-360 should plan around it.

Gross Revenue Retention (GRR)

Recurring revenue kept from existing customers **excluding any expansion** — only the damage from churn and contraction. Mathematically caps at 100%. [6]

```
GRR = (Starting MRR - Contraction - Churn) / Starting MRR
```

GRR is the honest “how leaky is the bucket” number. **Always read GRR alongside NRR**: a few big upsells can prop NRR to look healthy while GRR quietly erodes. *115% NRR with 82% GRR = heavy churn masked*

by upsells. GRR ~90% is widely treated as table-stakes. [6][7]

NRR benchmarks by segment — read the SMB row first

Print-Flow-360’s peer group is the **SMB (<\$25K ACV)** row. The enterprise and mid-market figures below are included for context only — **they are NOT our target**. Holding a low-ACV self-serve print tool to 118%/108% NRR is the most damaging mis-application in this whole document.

Segment (by ACV)	NRR (median)	Confidence	Source
SMB (<\$25K) — OUR PEER GROUP	~97% (top quartile >105%)	Solid	Optifai N=939 (2026) + SaaS Capital 2025 agree exactly [5][6]
Mid-market (\$25K–\$100K)	~102–108% (rises with ACV)	Solid	SaaS Capital (\$25–50K = 102% median); Optifai (~108%) [5][6]
Enterprise (>\$100K)	~118% (top quartile >130%)	Solid	Optifai 2026; SaaS Capital 2025 (both =118%) [5][6]
All private SaaS (blended)	~101% now, down from ~105%+ peak (ZIRP era)	Solid	KeyBanc / SaaS Capital 2024–25 [6]
GRR “table stakes”	~90% (KeyBanc dipped to ~86% in 2023)	Solid	SaaS Capital; KeyBanc 16th annual survey [6]

Bessemer Good / Better / Best NRR tiers — 100% / 110% / 120%+ (State of the Cloud 2023). [14] Real and correctly attributed — **but this is an enterprise/expansion-heavy benchmark**. See the folklore section: applying it to low-ACV SMB is a category error.

4. SaaS Quick Ratio

Devised by Mamoon Hamid (Social Capital). A one-glance read on whether growth is efficient or you’re filling a leaky bucket. [8]

Quick Ratio = (New MRR + Expansion MRR) / (Churned MRR + Contraction MRR)

Metric	Value	Confidence	Source
Quick Ratio “healthy” bar	~4 (“add \$4 for every \$1 lost”)	Directional	Mamoon Hamid / Social Capital — VC rule-of-thumb, not an empirical median [8]

Caveat for an early team: very early companies post huge Quick Ratios simply because their churn base is tiny — the number is near-meaningless at small scale. **Do not over-index on it weekly** until the paying base is large enough for the denominator to mean something.

5. The hybrid-revenue problem: subscription MRR vs platform GMV / take-rate

This is the section that matters most for Print-Flow-360 specifically. We have two revenue types that obey different math:

Subscription revenue (recurring)

= customers × plan price

→ this is MRR/ARR

Marketplace/transaction revenue

= buyers × transactions × AOV × take-rate

Blending order/transaction revenue into MRR is the classic distortion. It makes MRR look huge and volatile, breaks every retention and Quick-Ratio calc, and misleads on valuation — recurring revenue earns far higher multiples than transactional revenue. **Track them on separate P&L lines and separate dashboards.** [12]

GMV and take-rate

GMV = total dollar value of orders transacted through the platform in a period

(NOT your revenue)

Platform transaction revenue = GMV × take-rate

GMV is a **scale / engagement signal and a leading indicator of customer health** — a store doing volume won't churn — but it must never be summed into MRR. [11][12]

Metric	Value	Confidence	Source / nuance
Marketplace take-rate (typical)	~10–30% of GMV	Directional	Reforge / take-rate guides [11]. Goods-vs-services nuance: physical-goods marketplaces run lower (~5–20%); services/exclusivity platforms run higher (up to 30%+).

Relevance caveat: take-rate is a *marketplace* metric. Print-Flow-360 is primarily a **SaaS subscription tool** — it may take 0% on orders, or only a payment-processing spread. Take-rate only matters if/when we add a marketplace or commission layer, and even then it must never be blended into MRR/ARR.

6. Committed vs recognized revenue & deferred revenue (annual prepay)

Bookings

= total signed contract value

Billings

= amount invoiced

Recognized

= service actually delivered (revenue you can report)

When a customer **prepays annually**, you cannot book it all as revenue on day one. Under accrual/GAAP the unearned portion sits as **deferred revenue (a liability)** and is recognized ~1/12 each month as service is delivered. [9][10]

On sale: Cash +annual amount, Deferred Revenue (liability) +annual amount
Each month: recognize (annual amount / 12) as revenue, reduce deferred revenue by the same
→ MRR contribution = annual / 12 throughout the term

Practical upshot for our team: annual plans boost cash and lower churn (the funnel doc already recommends the annual “two months free” framing) — but **don’t let the upfront cash masquerade as a single-month revenue spike**. Committed MRR (CMRR) and revenue backlog (RPO) capture contracted-but-not-yet-earned revenue if/when we need a forward view.

7. What this means for Print-Flow-360

Concrete, opinionated guidance for a low-ACV SMB print SaaS with a non-technical customer base and limited eng time.

Definitions to lock in (non-negotiable)

1. **MRR/ARR = subscription plan fees only** (Starter/Growth/Pro), normalized monthly; annual → /12. Exclude every print-order dollar, setup fee, one-time design charge, and usage overage. Blending order revenue breaks every downstream metric and overstates valuation.
2. **GMV and take-rate revenue live on a separate dashboard line** from subscription MRR. GMV is the engagement/health and leading-churn signal; report any payment-processing spread or platform fee as “transaction revenue = GMV × take-rate,” distinct from MRR.

What to instrument

3. **Emit subscription events cleanly, separate from order events**, each with before/after plan + amount: `subscription_started`, `plan_upgraded`, `plan_downgraded`, `subscription_canceled`, `subscription_reactivated`. This is what makes the waterfall, NRR, GRR, and Quick Ratio *computable*. Per our “silent-lie” guard: **a movement not captured is a diagnosis lost**.
4. **Build the MRR movement waterfall as the core weekly artifact** (New / Expansion / Reactivation / Contraction / Churn). For a low-ACV self-serve base you’ll likely see lots of small New + meaningful Churn + little Expansion — the waterfall makes that visible so you fix the right lever (almost certainly retention/activation).
5. **Watch Contraction MRR weekly as an early-warning churn signal**. Wire a save-motion (in-app message / founder outreach, consistent with the founder-led acquisition strategy) when a store downgrades or its GMV drops sharply.
6. **Instrument an `expansion_trigger` event when a store crosses a tier threshold** (order volume / product count). This is the engine for the one realistic expansion path below.

What to do strategically

7. **Engineer ONE expansion path that fits this audience: usage-based tier graduation**. As a store’s order volume / product count grows, nudge Starter → Growth → Pro. This converts platform success

(GMV growth) into expansion MRR — the only durable NRR lever for low-ACV SMB. Add-ons (design studio, B2B accounts) are secondary expansion levers.

8. **Expect and accept SMB NRR < 100% (~97% segment median).** Plan to grow primarily via new logos + modest expansion, not expansion alone. Set internal targets against the SMB row, not enterprise.
9. **Treat annual plans as a churn-reduction + cash lever, accounted correctly:** book as deferred revenue, recognize 1/12 per month; MRR contribution = annual/12, never a one-month spike.
10. **Reconcile the two revenue worlds for the board in ONE table but NEVER in one number:**

Line	What
(a) Subscription ARR	the multiple-bearing recurring base
(b) Trailing GMV	scale / health indicator
© Take-rate / transaction revenue	GMV × take-rate, if any
(d) Total revenue	(a) + ©, shown explicitly as a sum — never collapsed into “ARR”

What to track WEEKLY (short — limited eng time)

- (1) Net New MRR + the 5-part waterfall
- (2) Logo churn count + churned MRR
- (3) Trial → paid conversion (ties to the existing North Star)
- (4) GMV on its own line as the health / leading indicator

What to IGNORE / defer for now

- **Quick Ratio precision** — numbers too small to be meaningful early.
- **NRR/GRR cohort math at sub-scale** — compute *monthly*, not weekly, until you have a few hundred paying stores.
- **Blended “total revenue” vanity figures** presented as ARR.

What to A/B test

- Annual “two months free” framing vs monthly default (cash + churn impact).
- In-app **tier-graduation nudge** copy/timing when a store crosses a volume threshold (does it lift Expansion MRR?).
- **Reactivation** win-back offers to recently churned stores (cheaper than new acquisition?).
- Contraction **save-motion** (in-app vs founder outreach) when a downgrade or GMV drop fires.

8. Folklore & mis-attributed stats — flag these explicitly

These circulate as “facts” in SaaS-metrics folklore. Do not repeat them uncritically.

- **“120% NRR is the bar every SaaS should hit.”** Mis-applied. The 100/110/120% Good/Better/Best framing is real (Bessemer, State of the Cloud 2023) but is an **enterprise/expansion-heavy** benchmark. For low-ACV SMB like Print-Flow-360, **~97% NRR is the segment median** — holding ourselves to 120% is the wrong target and will mislead strategy. [14]
- **Quoting Snowflake-style NRR as an aspirational target.** Snowflake’s **dollar-based NRR was 158% at IPO (as of July 31, 2020, per the S-1)** and **peaked at ~169% in early 2021** — *169% is a later peak, not an alternate IPO figure*. It has since normalized toward ~125–130% (directional, FY2026) as the base matured. These are extreme usage-consumption, enterprise data-warehouse figures — **irrelevant to a self-serve SMB print tool**. [12]

Metric	Value	Confidence	Source
Snowflake NRR at IPO (Jul 31, 2020)	~158%	Verified	S-1 / public filings [12]
Snowflake NRR peak (early 2021)	~169%	Verified	Public filings [12]
Snowflake NRR normalized (FY2026)	~125–130%	Directional	Recent quarterly filings [12]

- **Treating ARR as “annual bookings” or “annual cash collected.”** ARR is strictly **MRR × 12**, a run-rate snapshot. Summing a year of cash (including one-time print revenue) and calling it ARR is the single most common and most damaging recurring-revenue misstatement. [13]
- **Blending GMV or transaction revenue into MRR/ARR.** Common in marketplace-flavored startups to inflate headline numbers; it destroys the meaning of every retention metric and is heavily discounted (or flagged as a red flag) by investors. [12]
- **“Quick Ratio of 4 is a hard rule.”** It’s a VC rule-of-thumb (Mamoon Hamid), not an empirically-derived median. Early-stage companies post huge Quick Ratios because their churn base is tiny — near-meaningless at small scale. [8]
- **“A 5% increase in retention boosts profits 25–95%.”** The *direction* (retention compounds) is sound, and the attribution to **Reichheld / Bain (The Loyalty Effect, 1996; original HBR work 1990)** is actually **correct** — the folklore is the **over-precision** and the frequent claim that it came from a single tidy HBR study. Use the direction, not the exact band.
- **Counting an annual prepayment as a single-month MRR/revenue spike.** It overstates that month and understates the rest; annual prepay is deferred revenue recognized 1/12 monthly, contributing annual/12 to MRR throughout the term. [9][10]
- **Including setup/onboarding/one-time design fees or usage overages in MRR “because it’s revenue.”** It is revenue, but not *recurring* revenue — including it inflates MRR and corrupts the retention and Quick-Ratio math. [1][13]

Sources

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(waterfall components) — <https://help.chartmogul.com/article/157-chart-net-mrr-movements> [3] ChartMogul Help Center — Understanding MRR movements — <https://help.chartmogul.com/article/163-understanding-mrr-movements> [4] ChartMogul — Exploring Expansion and Reactivation MRR — <https://chartmogul.com/blog/reactivation-expansion-mrr/> [5] Optifai — B2B SaaS Net Revenue Retention Benchmark (N=939, by segment/ACV) — <https://optif.ai/learn/questions/b2b-saas-net-revenue-retention-benchmark/> [6] SaaS Capital — What is a Good Retention Rate for a Private SaaS Company? (2025 survey) — <https://www.saas-capital.com/blog-posts/what-is-a-good-retention-rate-for-a-private-saas-company/> [7] (read alongside [6]) KeyBanc Capital Markets — 16th Annual Private SaaS Survey (GRR ~86% 2023 trough, trending ~90%) [8] Cobloom — SaaS Quick Ratio: definition, formula & benchmarks (Mamoon Hamid / Social Capital) — <https://www.cobloom.com/blog/saas-quick-ratio-how-to-measure-your-startups-revenue-health> · Jonathan Hsu (Social Capital), Accounting for Revenue Growth — <https://medium.com/swlh/diligence-at-social-capital-part-2-accounting-for-revenue-growth-551fa07dd972> [9] The SaaS CFO — Guide to Deferred Revenue and SaaS Revenue Recognition — <https://www.thesaascfo.com/deferred-revenue-saas/> [10] Chargebee — Ultimate Guide to SaaS Revenue Recognition (bookings/billings/revenue, deferred) — <https://www.chargebee.com/resources/guides/saas-revenue-recognition-guide/> [11] Reforge — The 8 Most Important Metrics for Marketplace Growth (GMV, take-rate) — <https://www.reforge.com/blog/brief-the-8-most-important-metrics-for-marketplace-growth> · Alexander Jarvis — What Is GMV in SaaS — <https://www.alexanderjarvis.com/what-is-gross-merchandise-value-in-saas/> [12] Software Equity Group — How Net Revenue Retention Impacts SaaS Valuation (Bessemer tiers, Snowflake) — <https://softwareequity.com/blog/net-retention-public-saas-companies/> (Snowflake figures cross-referenced against the Snowflake S-1 / public filings) [13] Wall Street Prep — Monthly Recurring Revenue (MRR): Formula + Calculator (exclusions) — <https://www.wallstreetprep.com/knowledge/monthly-recurring-revenue-mrr/> [14] Bessemer Venture Partners — State of the Cloud 2023 (Good/Better/Best NRR tiers) [15] DigitalApplied — Net Revenue Retention Benchmarks 2026 (vendor-blog; treat % as directional) — <https://www.digitalapplied.com/blog/net-revenue-retention-benchmarks-2026-saas-expansion-data>

Related internal research

- [readme/CONVERSION_FUNNEL_RESEARCH_2026-06-15.md](#) — AARRR funnel, no-card 14-day reverse trial, North Star = store live + first order in 7 days.
- [readme/ACQUISITION_CHANNELS_2026-06-15.md](#) — founder-led outreach + print communities (primary); BOFU SEO + free design tool (secondary).

SaaS Unit Economics: CAC, LTV, CAC Payback, Magic Number, Rule of 40

Source: [readme/research/metrics-analytics/METRICS_SAAS_UNIT_ECONOMICS_2026-06-16.md](#)

IN SHORT — KEY TAKEAWAYS

This document explains the key numbers a software business uses to know if it is making money per customer — and which ones actually matter for a small, early-stage business like this one. It matters because getting these numbers wrong leads to bad decisions: spending too much to get customers, or thinking the business is healthy when it is not.

The main parts explained simply:

- **How much it costs to get one customer (CAC)** — This is every rupee/dollar you spend to bring in one new store owner: your time, tools, writing, ads, everything. The document warns that many businesses lie to themselves by only counting ad spend and forgetting to count their own working hours. The honest number is often 2–4 times bigger than it first looks.
- **How much a customer is worth over their life (LTV)** — This is the total profit you expect to earn from one customer before they leave. The document explains that this number is very easy to get wrong, especially when customers leave quickly (as small businesses often do). Treat it as a rough guess, not a fact.
- **The 3:1 rule (LTV vs cost to get a customer)** — A popular saying is that a customer should be worth at least 3 times what it cost to get them. The document explains this rule was made for big, mature companies and is often misused by early startups. Do not use it as a pass/fail test at an early stage.
- **How many months until a customer pays you back (CAC Payback)** — This is the most useful number for a small, cash-tight business. It simply asks: how many months of profit from this customer does it take to cover what I spent to get them? Under 12 months is healthy; over 18 months is a warning sign. Always check whether customers are actually staying that long.
- **Magic Number and Rule of 40** — Two metrics used by large, well-funded companies to measure growth vs. profit. The document is clear: ignore these completely until the business reaches roughly \$5M+ in annual revenue. Building reports for them now is wasted time.
- **Common myths to ignore** — Several widely repeated rules are simply wrong or only apply to big companies. The document lists them so you can spot and dismiss them when they appear in articles or pitch decks.

What to do with this: Track one number first — how many months it takes a new store to pay back what it cost to get them. Make sure that number includes your own time, not just cash spent. Everything else can wait.

Researched 2026-06-16 via a multi-agent workflow (research + adversarial fact-check pass). This is an internal grounding doc, not gospel. Every vendor-blog percentage and benchmark band below is a

hypothesis to instrument and A/B-test against our own cohorts, not a guarantee. Where a figure comes from a single vendor cohort study with undisclosed methodology, it is labelled as such inline — do not anchor pricing or fundraising on a borrowed number. Several widely-quoted “rules” in SaaS-metrics folklore are misattributed or misapplied; those are flagged explicitly in the [Folklore warnings](#) section.

Executive Summary

Headline recommendation: For a cash-constrained, low-ACV, self-serve SMB print SaaS, make **CAC Payback Period (in months)**, *margin-adjusted* the single headline economic metric — not LTV, not LTV:CAC, and definitely not Rule of 40. Compute a *fully-loaded* CAC (including founder/marketing time and the free design tool’s serving cost), instrument *per-tenant gross margin*, and read **payback alongside cohort churn**. Ignore Rule of 40 / Rule of X / Magic Number until you have a **repeatable, separately-measured S&M spend and ~\$5M+ ARR**.

Why this order, for *our* situation:

1. **CAC Payback needs no long-horizon churn projection.** It just answers “how many months until a new store pays us back from gross margin.” That makes it the most trustworthy unit-economics number for a high-churn, cash-tight SMB business. LTV, by contrast, extrapolates years of revenue we haven’t earned.
2. **LTV from the textbook $(ARPA \times GM) / churn$ formula will lie at SMB churn levels.** The $1/churn$ term is hyper-sensitive near high churn, and real SMB churn is front-loaded and non-constant. Treat any single-number LTV with suspicion; quote it with a confidence band built from observed cohort retention curves, or skip it.
3. **The famous “3:1 LTV:CAC rule” was derived from mature, steady-state public SaaS.** Applying it as a go/no-go gate pre-PMF (as nearly every seed deck does) is a misuse. A pretty 3:1 can sit on top of negative EBITDA and a fantasy 20-month “life.”
4. **Rule of 40, Rule of X, and Magic Number are scaled-company / repeatable-spend metrics.** They are volatile and not decision-useful at our stage. Building dashboards for them now is wasted eng time.

The cheap, correct build for our limited eng time: **one weekly spreadsheet/cron rollup** — new stores, fully-loaded CAC (including founder hours at a real rate), contribution margin per store, median CAC-payback-months by signup cohort, and % of stores activated within 7 days. Reuse the funnel events already specified in [CONVERSION_FUNNEL_RESEARCH_2026-06-15.md](#) (`card_added` , `trial_upgraded` , `first_order_received`) as the revenue/activation inputs. Don’t build a metrics warehouse yet.

1. CAC — Customer Acquisition Cost (three flavors, and which one is honest)

CAC is total sales+marketing cost divided by new customers acquired in the same period. There are **three distinct versions that get conflated constantly**, and the difference matters more for us than for a paid-heavy company.

Flavor	What it includes	When it's right	The trap
Fully-loaded CAC	S&M salaries + commissions + payroll tax + ad spend + tooling/software + content/creative + agency fees + events + allocated overhead	Unit economics. This is the number boards under-report and the one we should track.	Tedious to compile; requires allocating founder/marketing time
Blended CAC	Organic + paid customers averaged together	Quick top-line health check	Hides the true cost of paid programs; looks artificially low when growth is word-of-mouth/organic
Paid CAC	Only paid-channel customers ÷ only paid spend	Tells you whether paid actually works	N/A as a unit-economics number on its own

```
Fully-loaded CAC =  
(S&M salaries + commissions + payroll tax + ad spend + tooling/software  
+ content/creative + agency fees + events + allocated overhead)  
÷ new customers acquired in the period
```

Common miscounts to avoid:

- **Quoting ad-platform CPA as “CAC.”** This is the most common and most dangerous piece of SaaS-metric folklore. True fully-loaded CAC is commonly **~2–4× the ad-platform-reported number** once salaries, tooling, creative, and overhead are added [13][14]. Some analyses put the gap even higher (2–10× when CPA is mislabeled), so 2–4× is a *conservative, defensible* central estimate.
- **Excluding internal salaries / founder time.** For a self-serve PLG-ish motion, “S&M salaries” are largely **marketing + growth-eng + founder time**, not a sales team. If you leave founder hours out, CAC looks tiny — that’s the blended/organic miscount in disguise.
- **Time-period misalignment.** Don’t divide this month’s spend by this month’s customers when activation lags; match the period to the sales/activation cycle.

For Print-Flow-360: Per [ACQUISITION_CHANNELS_2026-06-15.md](#) the motion is founder-led outreach + print communities + BOFU SEO + a free design tool (paid search/affiliates skipped). So our CAC is **almost entirely unpaid labor and serving cost**, not ad spend. The honest CAC must include founder/marketing hours (allocated at a real hourly rate), the free design tool’s per-use serving cost, content/SEO effort, and tooling. A “CAC looks tiny because spend is near-zero” reading is the classic organic miscount — fix it by allocating time.

2. LTV — Lifetime Value (gross-margin-adjusted, and why it's fragile)

LTV must be **margin-adjusted**, never on revenue. The standard SaaS shortcut:

```
LTV = (ARPA × Gross Margin %) / Churn Rate
      where ARPA = avg recurring revenue per account,
            Gross Margin reflects cost-to-serve,
            1 / Churn = expected customer lifetime
```

Gross margin must subtract real **cost-to-serve**: hosting, S3 storage, PDF/file processing, support, payment fees. Revenue-only LTV systematically overstates value — and Print-Flow-360 has genuine per-tenant serving cost, so the overstatement would be material for us.

Why churn-based LTV breaks at SMB churn levels (the SMB trap)

The `1/churn` formula assumes a **constant churn rate forever**, which is false and dangerous at SMB churn:

- **Hyper-sensitive near high churn.** At low churn the curve is gentle; as churn rises, the `1/churn` term explodes/collapses. A small churn mis-estimate swings LTV by *years* of revenue.
- **Real churn is front-loaded and non-constant.** Early cohorts churn faster; a single blended rate misprojects.
- **It projects revenue you haven't earned and may never see.** With SMB monthly churn often 3–7%, you're extrapolating a 14–30 month “life” from customers who may not last a quarter.

Practitioners (David Skok, ChartMogul, The SaaS CFO) are explicit that **LTV is a projection, not actuals** [3][6]. Prefer **cohort-observed retention curves**. This fragility is precisely why **CAC payback is safer than LTV** for cash-constrained, high-churn SMB SaaS.

3. LTV:CAC ratio and the “3:1 rule”

```
LTV:CAC = LTV / fully-loaded CAC
          (both inputs MUST use the same margin and time basis)
```

Rule of thumb: ~3:1 “healthy,” <1:1 loses money per customer, >5:1 may signal under-investment in growth (per some investors — see caveat). It **misleads** when:

1. **LTV is a fragile high-churn projection** (garbage in) — a great-looking 3:1 built on a fantasy 20-month life.
2. **It ignores time/cash** — 3:1 with a 30-month payback can still bankrupt a cash-constrained startup.
3. **It's applied pre-PMF/seed** — the rule was derived from *mature public SaaS at steady state*. A 3:1 ratio can coexist with negative EBITDA if S&M efficiency erodes or churn spikes.

Sourcing caveat: The “>5:1 = under-investing” interpretation is *not* supported by The SaaS CFO “Ratio of Three” article (which makes no such claim and states no median) [3]. The “>5:1 excellent” *threshold* appears in vendor sources (Optifai/Prospero) [1][14]; the “under-investing” *reading* is commonly attributed to David Skok / a16z. Treat it as “may signal under-investment per some investors,” not a law.

4. CAC Payback Period — the SMB north-star economic metric

```
CAC Payback (months) = CAC / (ARPA_monthly × Gross Margin %)
                    = S&M spend / (New MRR × Gross Margin %)
```

Unlike LTV, **CAC payback requires no long-horizon churn projection** — it only measures how fast cash comes back. That makes it the **most trustworthy unit-economics metric for cash-constrained, high-churn SMB SaaS**, and the one we should put front-and-center.

Always read it alongside churn. A 10-month payback is fine if customers stay 3+ years but **lethal if many churn before month 12**. A payback longer than the average customer lifetime means you *never* recover the cost. Pair every payback chart with the matching cohort churn.

5. Magic Number — sales/GTM efficiency (company-level)

```
Magic Number = (Current-Qtr ARR - Prior-Qtr ARR) / Prior-Qtr S&M spend
variant: = (×4 of quarterly net-new REVENUE) / Prior-Qtr S&M (annualized)
```

It answers: *how many dollars of new annual recurring revenue does each S&M dollar buy?* It’s essentially the inverse cousin of CAC payback at the company level. **Origin:** Rory O’Driscoll / Scale Venture Partners (from analyzing Omniture — “>\$2 first-year revenue per \$1 of GTM,” the “It’s Magic!” piece); the 0.75 threshold was authored and popularized by **Lars Leckie** (2008) [8][9].

Two formula variants exist — net-new *ARR* / prior-qtr S&M vs. ×4 of quarterly net-new *revenue* / prior-qtr S&M. They produce different absolute numbers, so **a quoted “0.75” is meaningless without knowing the variant**. Confirm which a benchmark uses before comparing.

Magic Number is **most meaningful once you have a repeatable, separately-measured S&M spend** — i.e. NOT at pre-revenue seed, and NOT when “spend” is mostly founder time. Defer it.

6. Rule of 40 — growth + profitability balance

Rule of 40 = ARR/MRR Growth Rate % + Profit Margin % ≥ 40%
Profit term = EBITDA margin | FCF margin | operating margin (STATE which)

A **trade-off rule**: high-growth/unprofitable or slow-growth/profitable can both “pass.” Popularized by **Brad Feld (~2015, building on Fred Wilson)** [10]. The profit term has variants — EBITDA (most common; strips interest/tax/D&A), FCF (growth investors increasingly prefer; captures real cash after capex/SBC), or operating margin. **State which you use** — EBITDA vs FCF differ materially with stock-based comp.

Bessemer’s “Rule of X” is the modern refinement, weighting growth ~2–3× because growth compounds while margin is a one-period benefit:

Rule of X = (Growth Rate × ~2) + FCF Margin (multiplier ~2 private, ~2-3 public)

When Rule of 40 barely applies to an early product

Rule of 40 is built for **mature/scaled SaaS**. At seed/early stage it is volatile and misleading: growth off a tiny base is a huge percentage, and margins are deeply negative *by design* (you’re buying PMF and market share). Scale VP is blunt: “Rule of 40 Does Not Compute for Early-Stage Startups.” For an early SMB print SaaS the right early metrics are **activation/time-to-value, CAC payback, gross margin, and net dollar/logo retention** — not Rule of 40.

Benchmark tables

Confidence legend: **Verified** = confirmed against the primary source · **Solid** = confirmed against a credible benchmark dataset · **Directional** = single-vendor cohort / undisclosed methodology, use as a hint only · **Fabricated-avoid** = folklore, do not cite as fact.

CAC Payback by segment

Metric	Value	Confidence	Source
CAC Payback — median across all B2B SaaS (the anchor)	~ 15 months	Solid	Benchmarkit 2025 [6]
CAC Payback — SMB / low-ACV (<\$15K ACV)	~8–12 months	Solid	Benchmarkit 2025; Optifai (N=939) [6][2]
CAC Payback — self-serve / fast-activating SMB (industry-specific)	~1–7 months — floor, not a target; self-selected 50+ co sample	Directional	First Page Sage [5]
CAC Payback — Mid-market (\$15K–\$100K ACV)	~14–18 months (band is <i>not</i> strictly monotonic with ACV; \$10–50K can cost more than \$50–100K)	Solid	Optifai/Benchmarkit 2025 [2][6]
CAC Payback — Enterprise (>\$100K ACV)	~18–24 months (worst industry ~30–31 mo, e.g. Retail)	Solid	Optifai/Benchmarkit; First Page Sage [2][6][5]
CAC Payback — best-in-class	under 12 months (efficient/self-serve ~5–7 mo — label as self-serve, not a universal norm)	Solid	Benchmarkit 2025 [6]

LTV / LTV:CAC

Metric	Value	Confidence	Source
LTV:CAC — healthy minimum	~3:1	Directional	Optifai (N=939, 2026, undisclosed methodology, self-selected) [1]
LTV:CAC — median	~3.2:1	Directional	Optifai (same caveat) [1]
LTV:CAC — “>5:1 excellent / may signal under-investment”	>5:1	Directional	Optifai/Prospero for threshold; “under-investing” reading = Skok/a16z, not SaaS CFO [1][14] [3]
SMB LTV (absolute, illustrative)	~\$15K–\$40K over a 2–3 yr life — single-vendor cohort; never anchor pricing/fundraising on it	Directional	Optifai segment study [1]
Fully-loaded CAC vs platform-reported CPA gap	~2–4× (conservative; some sources 2–10×)	Directional	CAC composition guides [13][14]

GTM efficiency / company-level (scaled companies)

Metric	Value	Confidence	Source
Magic Number thresholds	<0.75 fix GTM before scaling · >0.75 “pour on gas” · >1.5 “exceptional” (secondary, not in Leckie’s original 0.75 quote)	Solid (0.75) / Directional (1.5)	Scale VP (O’Driscoll origin); Leckie [8][9]
Rule of 40 — avg vs top decile (scaled <i>public</i> SaaS)	avg BVP Cloud Index ~31%; top decile ~48% (late 2023)	Solid	Bessemer Cloud Index [11]
Rule of X — avg vs top (Bessemer)	avg ~50%; top decile ~80%; good/better/best ≈ 60 / 65 / 80% (growth weighted ~2.3×)	Solid	Bessemer “The Rule of X” [11]
Rule of 40 applicability floor	~\$5M+ ARR at minimum; most reliable at ~\$15–20M+ ARR or scaled/public — volatile/misleading at seed	Solid	Wall Street Prep; Software Equity Group; Bessemer; Scale VP [10][12][11]

What this means for Print-Flow-360

Concrete, prioritized guidance for a low-ACV SMB print SaaS with limited eng time.

Instrument first (in this order of urgency)

1. **CAC Payback Period (months), margin-adjusted — the headline.** Put it on the founder dashboard. It needs no fragile churn projection and directly answers “how fast does a new store pay us back.”
2. **Fully-loaded CAC — including founder/marketing time.** Do *not* use ad-platform CPA. Allocate founder hours at a real hourly rate; include the free design tool’s serving cost, content/SEO effort, and tooling. A “tiny CAC” reading from mostly-unpaid labor is the blended/organic miscount — fix it.
3. **Per-tenant gross margin (cost-to-serve).** Add a simple per-store estimate: `revenue - payment fees - allocated infra/PDF-service/S3/support cost` . Payback and LTV are only honest on **contribution margin**, not revenue. This is the hinge for everything else.
4. **Cohort-observed revenue-retention curves**, not `1/churn` LTV. Extend the Retention dashboard from `CONVERSION_FUNNEL_RESEARCH_2026-06-15.md` to plot **\$ retained per signup-month cohort**. Quote LTV with a confidence band, never a single hero number.
5. **CAC Payback segmented by activation.** The funnel North Star is “stores with ≥1 order in last 7 days,” activation = “store live + first order in 7 days.” Activation is the *leading indicator* of payback — an activated store contributes margin sooner. Report payback split by **activated-in-7-days vs not**, to prove the activation investment’s economic payoff.

What to ignore early (vanity / too-early for us)

- **Rule of 40 and Rule of X** — scaled-company (~\$5M+, ideally \$15–20M+/public) metrics; volatile and meaningless at our stage. Don’t build dashboards for them.

- **Absolute LTV hero numbers** and the “**3:1 LTV:CAC**” as a **go/no-go gate** while pre-PMF.
- **Magic Number** — premature until there’s a repeatable, *separately-measured* S&M spend (not founder time). Revisit once paid/repeatable acquisition exists.
- **Borrowed absolute dollar benchmarks** (e.g. “SMB LTV is \$X”) from vendor cohort studies — directional at best; never anchor pricing or a raise on them.

What to A/B test (on our own funnel)

- **No-card reverse trial vs card-required** (the no-card 14-day reverse trial is recommended in `CONVERSION_FUNNEL_RESEARCH_2026-06-15.md`). It lowers signup friction but may *lengthen* payback by admitting more free riders. **Test it on CAC PAYBACK PER COHORT, not just signups/trial-to-paid** — the no-card variant only wins if its larger paid base still pays back in a comparable number of months.

Targets — set segment-appropriate ones, don’t borrow

- As a self-serve SMB tool, aim for **CAC payback well under 12 months** (self-serve SMB can run ~1–7 months *when activation is fast* — but that floor is industry-specific, **not** a number you’ve “failed” to hit; the all-B2B median is ~15 months).
- Treat **>12 months** as a signal to fix activation/margin/pricing; **>18 months** as a structural GTM problem.
- **Always pair the payback chart with matching cohort churn**, so a “good” payback isn’t hiding customers who leave before they pay back.

The cheap build (limited eng time)

A **single weekly spreadsheet or cron rollout** is enough — do not build a metrics warehouse yet:

- new stores this week,
- fully-loaded CAC (incl. founder time at a real rate),
- contribution margin per store,
- median CAC-payback-months by signup cohort,
- % of stores activated within 7 days.

Reuse the funnel events already specified (`card_added` , `trial_upgraded` , `first_order_received`) as the revenue/activation inputs.

Folklore warnings (mis-attributed or fabricated stats)

Flag these explicitly whenever they come up in decks, vendor blogs, or internal debate:

1. **The “3:1 LTV:CAC rule” is NOT a universal law.** Popularized by David Skok (~2010) from **mature, steady-state public SaaS** (HubSpot, Salesforce, NetSuite) with stable churn and sub-12-month payback. Applying it to seed/pre-PMF SMB (as nearly every seed deck does) is a misuse. A 3:1 ratio can sit on top of negative EBITDA and high churn.

2. **1/churn LTV quoted as a hard number is folklore.** It's a fragile projection; at high SMB churn the term is hyper-sensitive, so a small churn mis-estimate swings "LTV" by years of revenue. Demand the underlying churn and cohort curve behind any single-number LTV.
3. **Platform-reported CPA quoted as "CAC" is the most common and most dangerous SaaS-metric folklore.** True fully-loaded CAC is typically **2–4×** higher (sometimes more) once salaries, tooling, creative, and overhead are added. Blended CAC similarly flatters the number by hiding paid-channel cost.
4. **Rule of 40 mis-applied to early-stage startups.** It's a scaled-company metric (~\$5M+ ARR, most reliable \$15–20M+/public). Seed-stage Rule-of-40 figures are volatile and not decision-useful. Bessemer themselves argue the equal-weight version is incomplete (their Rule of X weights growth ~2–3×).
5. **Magic Number "0.75" is meaningless without the formula variant.** Two variants circulate (net-new ARR / prior-qtr S&M vs. $\times 4$ net-new *revenue* / prior-qtr S&M) and produce different absolute numbers. Origin is Rory O'Driscoll / Scale VP (Omniture); the 0.75 threshold is Lars Leckie — *not* an anonymous "industry standard." The ">1.5 exceptional" tier is a secondary add-on, not in Leckie's original quote.
6. **Single absolute LTV/CAC dollar benchmarks** (e.g. "SMB LTV is \$X") come from vendor cohort studies (Optifai N=939, First Page Sage 50+ cos) with **self-selected samples and undisclosed methodology** — directional at best, never authoritative. Never anchor pricing or fundraising on a borrowed absolute.
7. **CAC-payback tables showing SMB at "1–7 months"** describe **self-serve, fast-activating** products and specific industries. Do not treat the floor as a target you've failed to hit; the all-B2B median is ~15 months.

Verifier note: this set contained no fabricated figures and no smuggled-in folklore (no misattributed "5% retention → 95% profit" Bain claim, no fabricated retention couplings). The core "Solid" benchmarks were confirmed against primary sources ([benchmarkit.ai](https://www.benchmarkit.ai), [bvp.com](https://www.bvp.com), [scalevp.com](https://www.scalevp.com)).

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